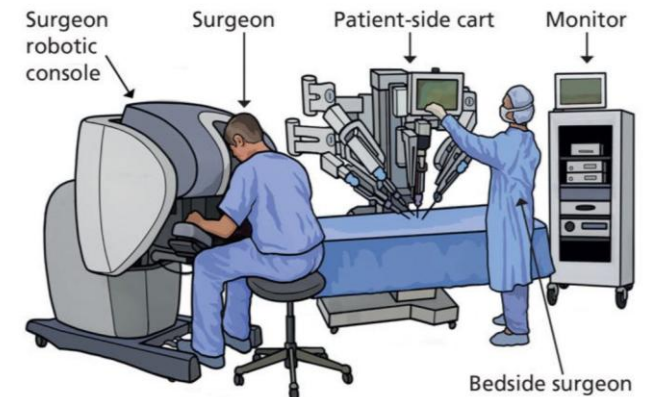


Robotic assisted surgery

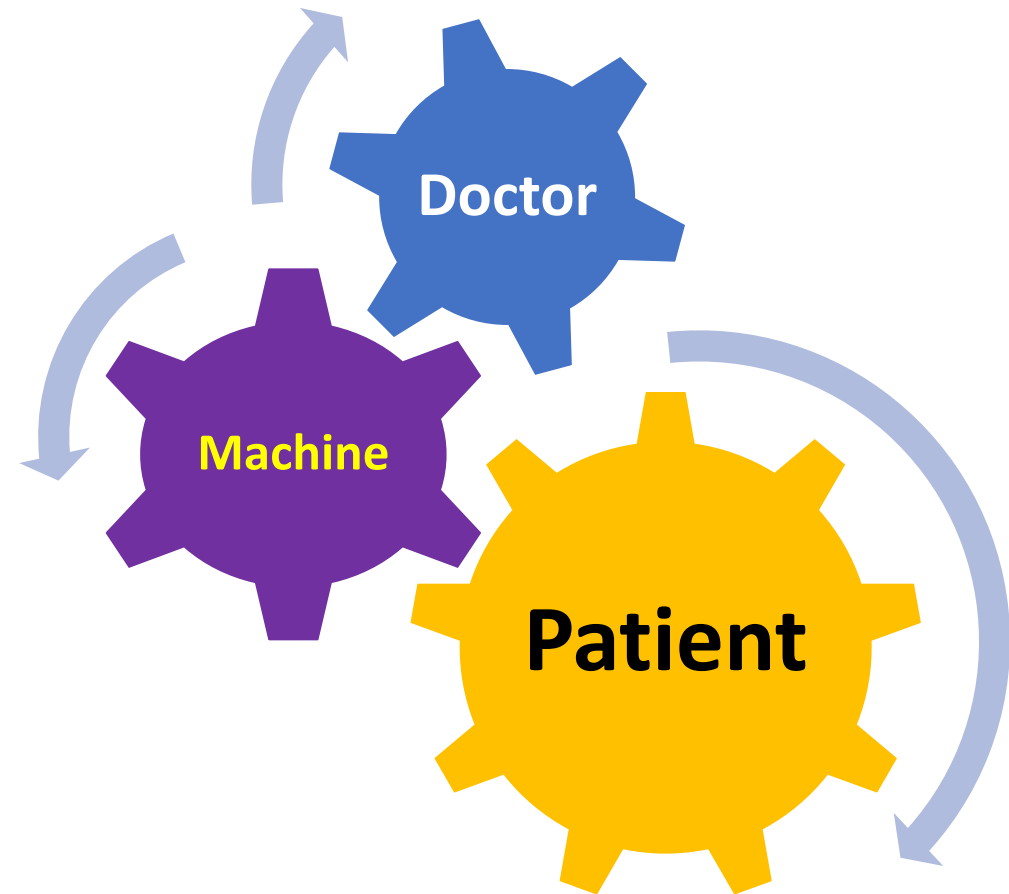
Introduction and principal

Boonlert Viriyapak



What?...All about “Biomedical Engineering” aimed to

- Perform fine surgical maneuver instead of human
- And develop telepresence surgical system
- Integrated technology to coordinate.....



Why?...Robotic benefits

- **Enhanced Precision:** Minimizes damage to surrounding tissues.
- **Improved Visualization:** Provides a 3D, magnified view of the surgical field.
- **Reduced Trauma:** Smaller incisions lead to less blood loss.
- **Faster Recovery:** Patients typically experience shorter stays and quicker recovery times.
- **Reduced Fatigue:** The surgeon can operate in a comfortable seated position and robotic arms provide steady support.



How?...

The surgeon controls robotic arms/instruments from a console

- With better 3D visualization, tremor filtering, and enhanced dexterity to achieve more precise surgery.
- It would make more convenient to control, lesser fatigue and more steady surgical dissection.
- In gynecological use, compared to traditional laparoscopic surgery, it was approved in safety and initially provided advanced technique to perform hysterectomy, myomectomy and sacro-colpopexy.
- Additional utilization in advanced/radical surgery for gynecologic malignancies with perioperative advantages while equal oncologic outcomes.

History and development



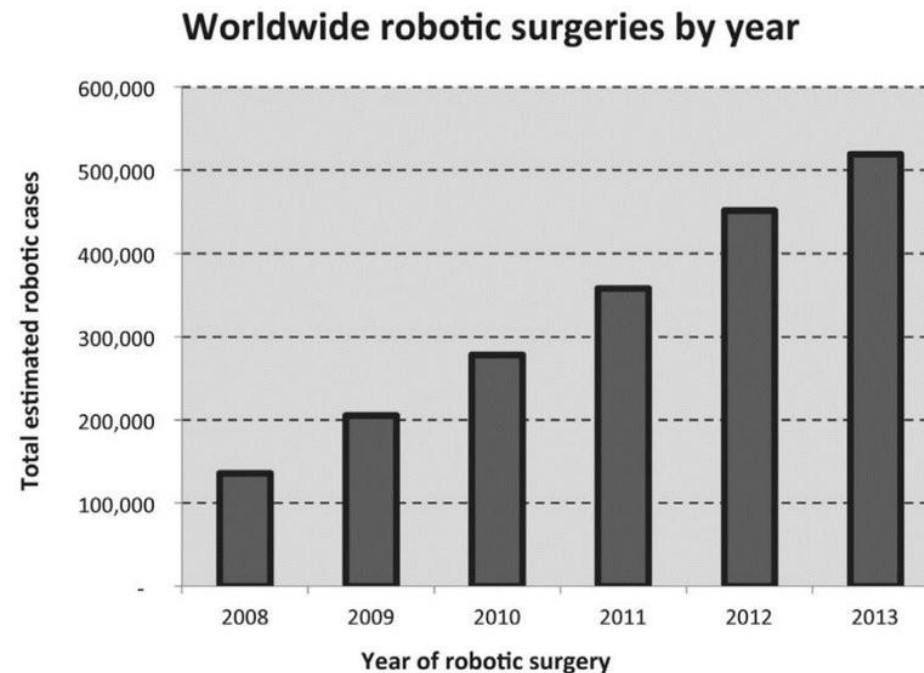
- In 1988, **the PUMA 560** was used to place a needle for a brain biopsy using CT guidance
- In 1987, **Stanford's** robotics was used in the first Laparoscopic cholecystectomy
- In 1992, **the PROBOT**, developed at Imperial College London was used to perform prostatic surgery

In 1997, **ZEUS** was the first robot to perform gynecological surgery.
It reconnected fallopian tubes in Ohio, USA.

The ZEUS robot was then purchased by **Intuitive Surgical, Inc.**



- In 2005, the US-FDA approved the use of the Da Vinci robotic system in gynecological surgery.
- From 2007-2013, the number of robotic surgical system over doubled in US (up to 2,000) and Europe. Overall approximated 1.5 million robotic surgeries had been performed that time.



Estimated number of robot-assisted cases worldwide. (Data from Intuitive Surgical investor presentation, 2014)

- The Thai FDA does not approve individual robotic surgery procedures, but rather the medical devices and systems used
- Robotic surgery, such as with the Da Vinci system, is widely used in Thailand for gynecological procedures like hysterectomies, ovarian cyst removal and early-stage cancers
- Greater precision, faster recovery and less pain compared to open Sx

Many types of robotic surgical system in uses

- 1. Supervisory controlled robotic surgery system.**
- 2. Tele surgical system.**
- 3. Da Vinci surgical system.**
4. Automated Endoscopic System For Optical Position (AESOP) robotic surgery system.
5. Shared control robotic surgery system.
6. Cyber Knife System

1. Supervisory controlled robotic surgery system

- Most automated type.
- There is a defined sequence of operations.
- Robot can't make adjustments, must be supervised.
- Mapping of patient in 3 ways: Planning, Registration, Navigation
- Used in hip and knee replacement procedures.



2. Tele surgery system

- Also called remote surgery.
- Surgery is performed by robotic system controlled by surgeon at distant site.
- Advantages:
 1. Assisting and training surgeons in developing countries
 2. Collaborating and mentoring during surgery by surgeons around the globe

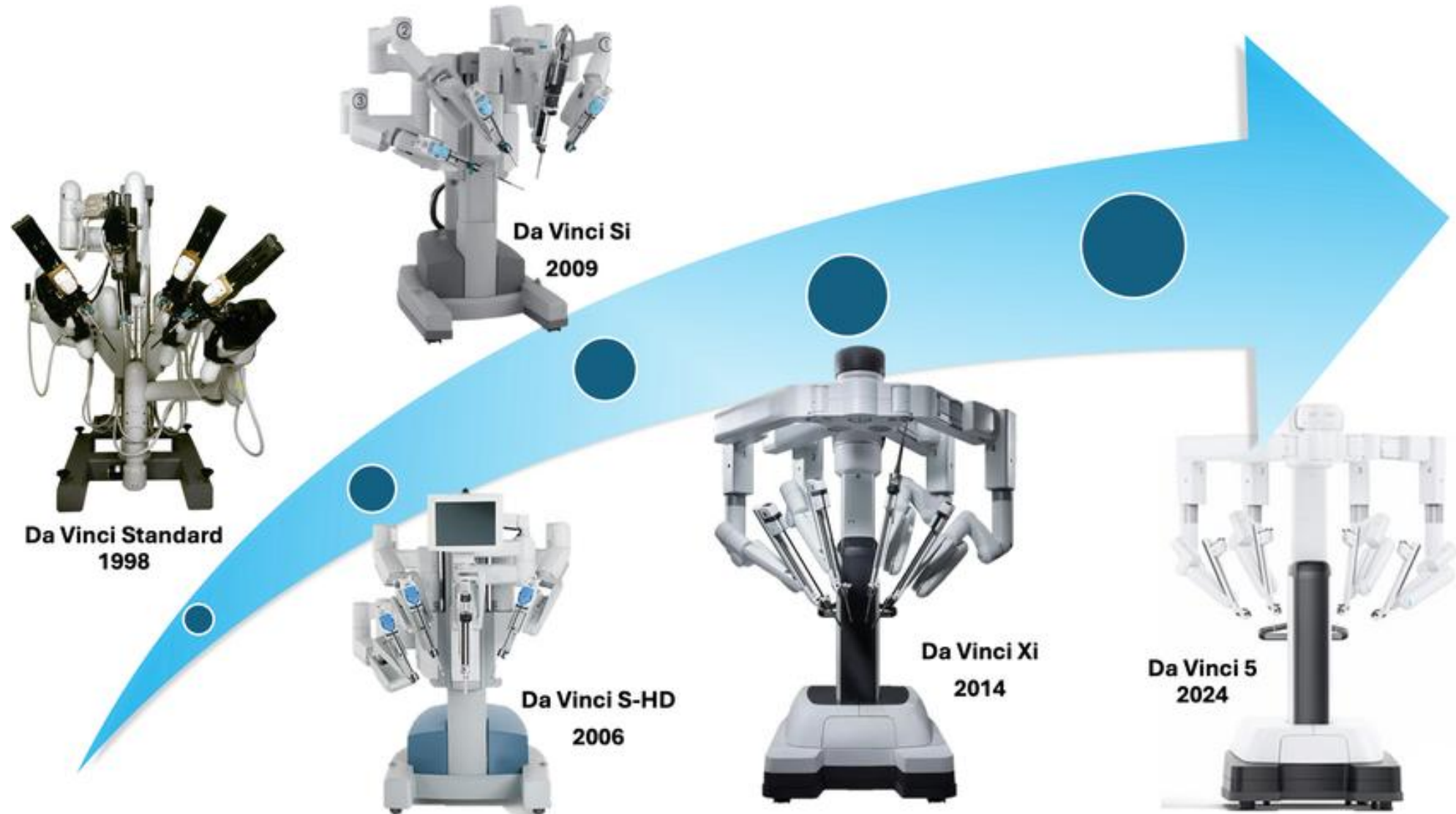


3. Da Vinci System

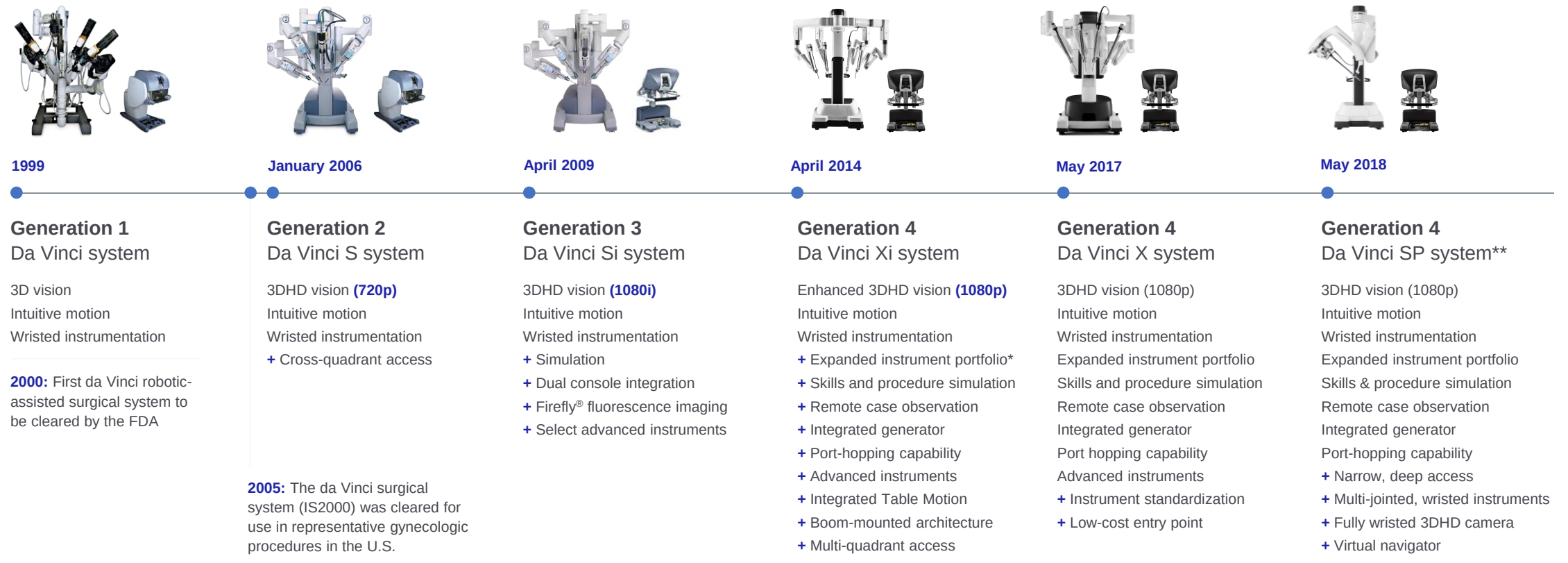
- Multiple robotic arms
- Camera control and end effector control



Evaluation of “Da Vinci” multiport robotic system



20 years of advanced robotic-assisted surgery learning and development



*Compared to the da Vinci Si system.

**The safety and effectiveness of this device for use in the performance of general laparoscopic surgery procedures have not been established. This device is only intended to be used for single-port urological procedures and for transoral otolaryngology surgical procedures in the oropharynx for benign tumors and malignant tumors classified as T1 and T2 with the da Vinci EndoWrist SP Instruments and the da Vinci SP surgical system (SP1098).

The da Vinci Xi system provides a multi-faceted solution



Fourth-generation standardization benefits

- Streamlined inventory management
- Consistent user interface and setup
- Platform solutions for different clinical and economic needs
- Broader suite of simulation modules
- Da Vinci OS4 intelligent software integration
- Increased capability through expanded product offerings

Capability and ease of use enhancements (Xi)

- Streamlined port placement (X, Xi)
- Full multi-quadrant access
- Automated setup tasks
- Advanced intraoperative adjustments
- Simplified cart positioning
- Optimized patient-side access

Advanced energy portfolio

Enhanced visualization and control



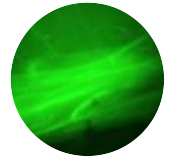
Full 30, 45, 60mm stapling portfolio



Simulation



E-100 Generator



Sensitive Firefly® (Xi)



Core Instruments



Integrated Energy



Vessel Sealer Extend



Scope Flip and Port Hopping



Integrated Table Motion (Xi)



Analytics and Advanced Software



SynchroSeal™



Endoscope Plus



Multi Quadrant Access (Xi)



IRIS Imaging



Force Bipolar w/ DualGrip



Handheld Camera



Da Vinci Energy

SynchroSeal

Streamline sealing.⁹ Optimize performance.¹⁰



Get enhanced efficiency¹¹ and versatility¹² from a single instrument with less than 2 seconds average seal time.¹³

With a single pedal press, SynchroSeal enables you to seal and cut vessels up to 5 mm in diameter and tissue bundles that fit in the jaws.

Vessel Sealer Extend

Performance you expect. Speed that you don't.¹²



Securely seal and cut vessels up to 7 mm in diameter or tissue bundles that fit in the jaws.

With an average sealing time of less than three seconds⁹, Vessel Sealer Extend may potentially improve efficiency during the surgical case.¹⁴

Powered by the E-100 generator

Notice a difference in speed.¹⁵



Seamlessly integrates into the da Vinci X and Xi surgical systems to power SynchroSeal and Vessel Sealer Extend⁹

Three energy modes: Coagulation, sealing, and transection⁹

When paired with SynchroSeal, enables a less than two seconds average seal time¹³

When paired with Vessel Sealer Extend, enables fast sealing time¹⁶ with low jaw temperatures¹⁷

⁹ Data on file.

¹⁰ Compared to HARMONIC ACE+7.

¹¹ Compared to Vessel Sealer Extend with ERBE VIO dV in terms of faster estimated workflow and seal activation times.

¹² Compared to Vessel Sealer Extend with ERBE VIO dV.

¹³ Mean seal time based on seal time from seal and sync mode.

¹⁴ Compared to Vessel Sealer Extend with ERBE VIO dV in terms of seal activation times.

¹⁵ Compared to ERBE VIO dV.

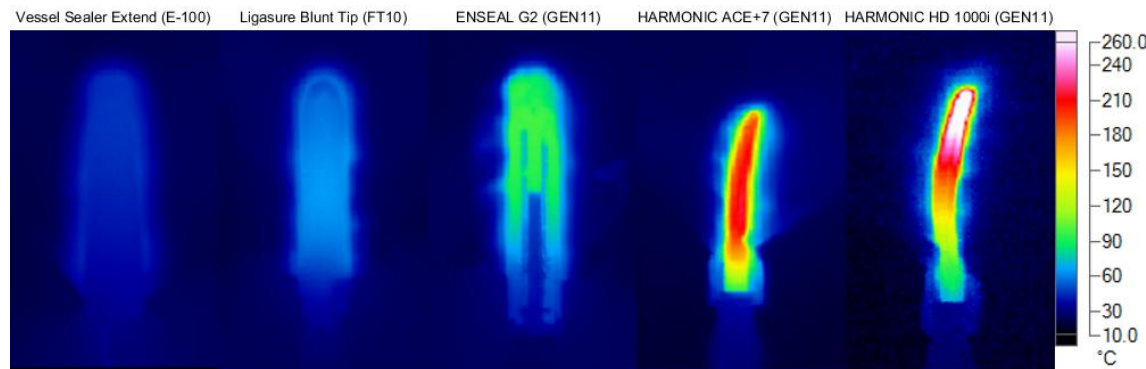
¹⁶ Compared to Vessel Sealer Extend with ERBE VIO dV in terms of seal activation times.

¹⁷ Compared to LigaSure Blunt Tip, ENSEAL G2 Articulating, HARMONIC ACE+7 and HARMONIC HD1000i. Data on file.

Vessel Sealer Extend from da Vinci Energy



- **Performance you expect. Speed that you don't.¹¹**
 - Compatible with E-100 generator, which enables a sealing time of less than 3 seconds⁹
 - 30% slimmer jaw profile¹⁸ for optimized dissection and tissue manipulation
 - Electrode extended by 3 mm¹⁸ for enhanced vascular access
 - Seals and cuts vessels up to 7 mm in diameter⁹ with 1–2 mm of lateral thermal spread⁹
 - Fully wristed with 60° of articulation⁹
- **Competitive jaw temperature comparison**



Operate at a cooler jaw temperature

When compared with competitive sealers, Vessel Sealer Extend shows the coolest peak temperature and the shortest cooling time.¹⁷



Keep your temperatures even lower

Vessel Sealer Extend operates at a cooler jaw temperature with E-100 than with ERBE VIO dV.¹¹

⁹ Data on file. ¹¹ Compared to Vessel Sealer Extend with ERBE VIO dV. ¹⁷ Compared to LigaSure Blunt Tip, ENSEAL G2 Articulating, HARMONIC ACE+7 and HARMONIC HD1000i. Data on file. ¹⁸ Compared to EndoWrist Vessel Sealer. Data on file.

Take advantage of a new level of grasping control*

- **Force Bipolar is available on the da Vinci X and Xi surgical systems**



DualGrip technology

Apply strong grip when needed



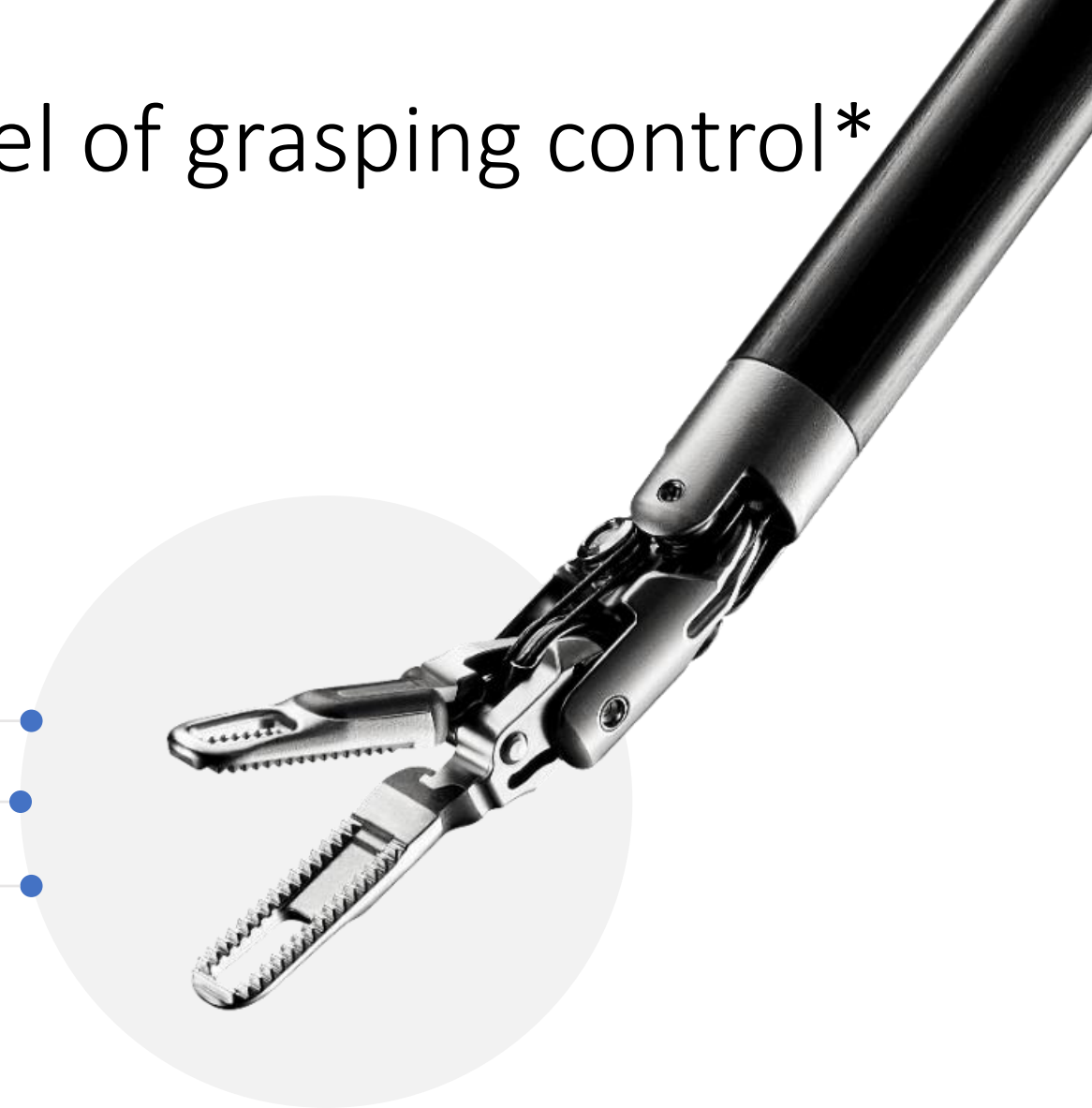
Bipolar functionality

Available in both modes



Efficiency-led innovation

Minimize instrument exchanges



*Compared to other da Vinci 8 mm instruments. Results from testing on file.



Da Vinci Components

Surgeon's Console

- Optimal hand-eye alignment
- Immersive 3D stereo viewer
- Comfortable seat posture – ergonomic
- Motion scaling & tremor reduction



Master controller controls the motion of the end effector wrist

Patient-side cart



A Da Vinci Robotic Column



EndoWrist instruments: 3–4 robotic arms, interchangeable end effectors

Imaging processing equipment and instruments

The Endoscope

- Fiber optic instrument
- Inserted through a small incision
- Has a very tiny video camera
- Shows the surgeon a 3-dimensional, magnified view
- Projects to a television screen



Electrosurgical system

- Power generation control
 - Monopolar
 - Bipolar
 - (Ultrasonic)



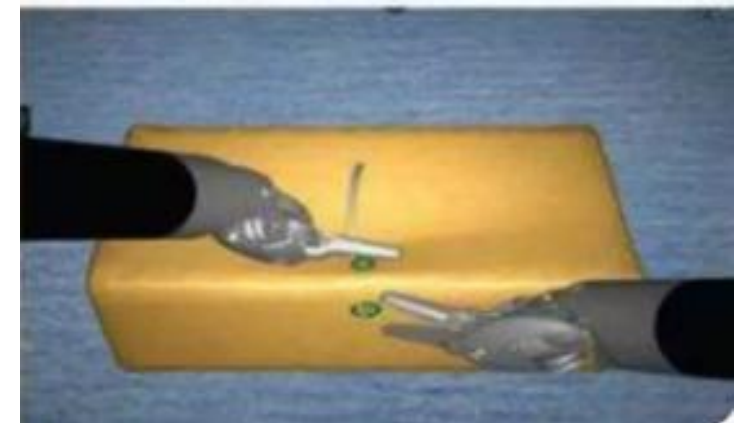
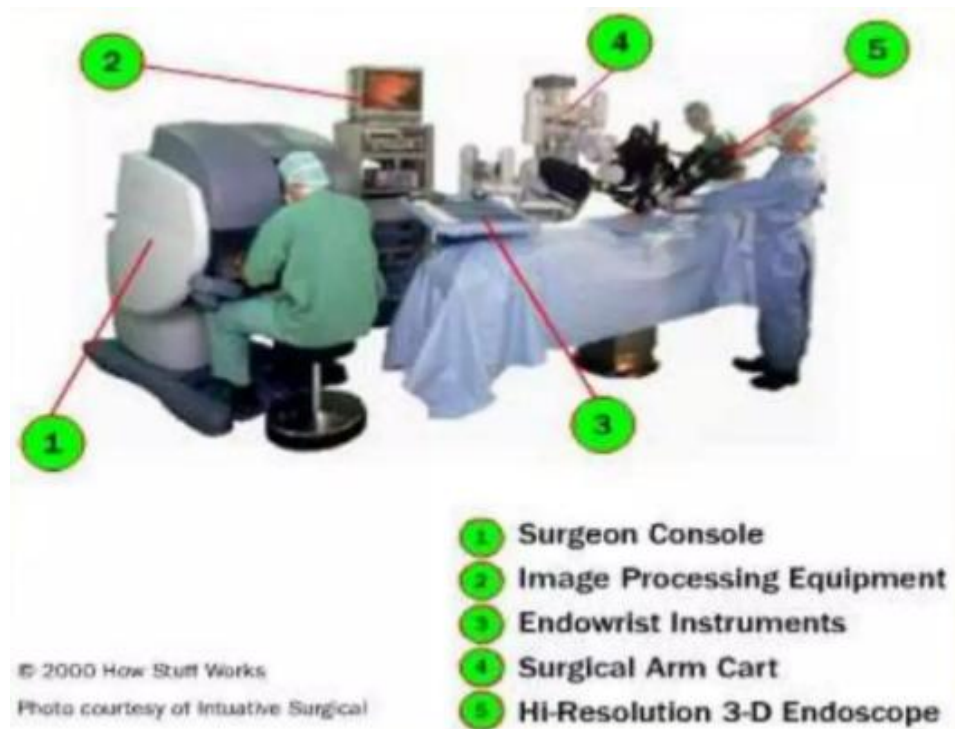
Next generation power generator

Specialty of Da Vinci

- Placing his fingers into the master controls, he is able to operate all four arms of the da Vinci simultaneously.
- Looking through a stereoscopic high-definition monitor that literally places inside the patient, giving him a better, more detailed 3-D view of the operating site than the human eye can provide.



- Wide array of instruments
- Quick instrument changes
- Rapid setup less than 15 minutes
- Visualization



Other competitors

Hugo Medtronic (US)



Asensus (US)

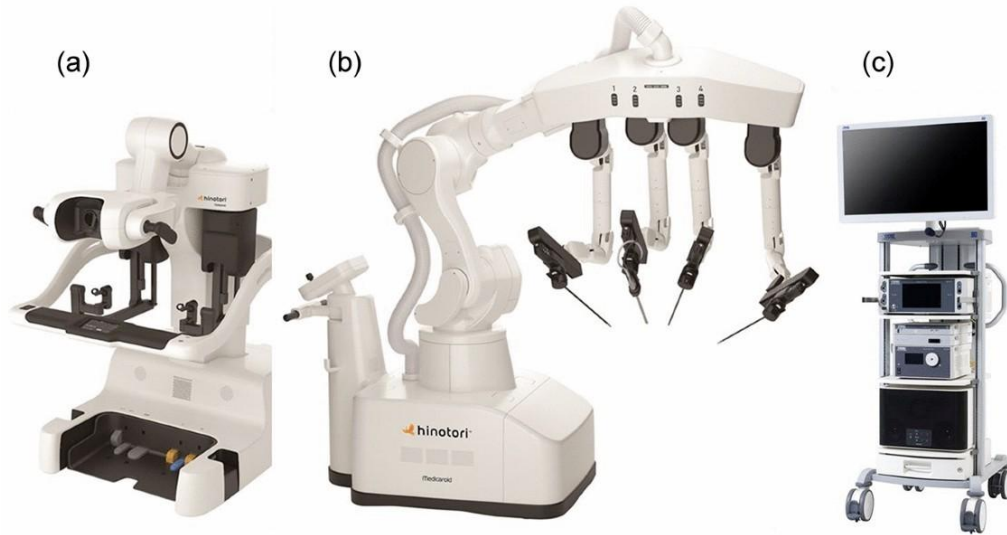


MicroPort MedBot (China)



Other competitors

Hinotori (Japan)

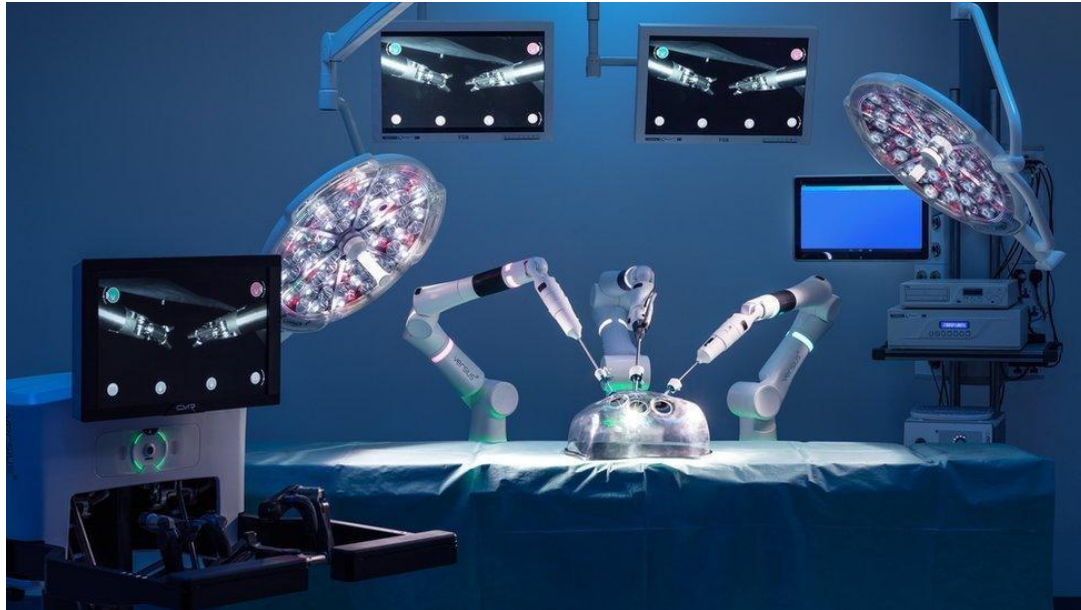


Revo-I (Korea)



Other competitors

Versius (UK)



Avatera (Germany)





Revo-i

Standalone

Mechanical RCM

Finger Gripper

Closed Console



Davinci

Standalone

Mechanical RCM

Finger Gripper

Closed Console



Versius

Modular

Software RCM

Joystick

Open Console



Hugo

Modular

Software RCM

Pistol type

Open Console

การใช้หุ่นยนต์ช่วยผ่าตัด (Robotic surgery)

- ระบบหุ่นยนต์ช่วยผ่าตัดระบบแรกที่มีการมาใช้ในการผ่าตัดทาง ศัลยศาสตร์ยูโรวิทยา คือ da Vinci Surgical System
- ถูกนำมาใช้ในการผ่าตัดทางศัลยศาสตร์ยูโรวิทยาครั้งแรกในประเทศไทย เมื่อวันที่ 16 กุมภาพันธ์ พ.ศ.2550 (February 16, 2007) ที่ โรงพยาบาลศิริราช โดยการนำของ รศ.นพ.ไชยรงค์ นวลยงค์ (ตำแหน่ง วิชาการในขณะนั้น) และคณะ
- หัตถการแรก คือ การผ่าตัดเอาต่อมลูกหมากออกทั้งหมด เพื่อรักษา โรคมะเร็งต่อมลูกหมาก เรียกว่า “Robotic Assisted Laparoscopic Radical Prostatectomy หรือ RALRP”



The 1st Successful Case of “da Vinci XI” Robot Surgery in Thailand!

🕒 Published 2019-07-23

For over 12-year of the initiation of the first public hospital who performed robotic surgery and more than 3,000 cases were succeeded. Division of Urology Surgery, Department of Surgery, Faculty of Medicine Siriraj Hospital has debuted the 3rd surgical robot with is the first model in Thailand (da Vinci XI). The 1st case has been successfully completed on July 23rd, 2019.

In the study of Yiengpruksawan et al. (2018) on Robotic Surgery in Thailand: Current Status and Future Development emphasizes “... any robotic surgery program in Thailand is a luxury – that is an undeniable fact. It is not, by any evidence-based information we have to date, superior to the existing technology.

However, the reality is that there is now no turning back on this technology on the local and international level. Moreover, it is no longer beyond the means of an increasingly affluent and internationalizing Thai medical community. But whatever goals we may set for robotic surgery in Thailand, the most important priority must be the safety of the patient and an excellent surgical outcome. These goals are definitely attainable through careful planning, the presence of an exemplary surgical team, ongoing self-evaluation, and analysis through prospectively collected data and videos of robotic surgeons, and last, but not least, a smart business plan...” (Siriraj Med J 2018; 70: 466–470).



Total Robotic Hysterectomy: Thailand's First Case Report of Gynecologic Robotic Surgery

Korakot Sirimai, M.D.*, Perapong Inthasorn, M.D.**, Pavit Sutthritpongsa, M.D.*, Prasong Tanmahasamut, M.D.***

Division of Gynecologic Endoscopy, **Division of Gynecologic Oncology, *Division of Reproductive Medicine, Department of Obstetrics and Gynecology, Faculty of Medicine Siriraj Hospital, Mahidol University, Bangkok 10700, Thailand.*

ABSTRACT

Objective: This study aimed to report the feasibility, trouble shooting and surgical technique in the total robotic hysterectomy for the patient with adenomyoma uteri.

Methods: A 51 year-old patient was diagnosed with adenomyosis. Total robotic hysterectomy was performed.

Results: Total operating time was 350 min, estimated blood loss was 50 ml, and length of hospitalization was 6 days. The pathologic section revealed adenomyosis with myoma uteri. The intraoperative and post-operative complications were unremarkable. The patient was in good conditions at 6th week, 3rd, 6th, 12th and 24th month.

Conclusion: Total robotic hysterectomy for benign gynecologic condition, such as adenomyosis, is safe and feasible. However, the sustained high consuming cost must be weighted with the patient's advantages.

Keywords: Robotic, hysterectomy, adenomyosis, myoma uteri

Siriraj Med J 2016;68:42-46

E-journal: <http://www.sirirajmedj.com/ojs>



Siriraj hospital

- “From 2007 – 2019, we have performed at least 2,500 patients with a successful result.”
- In 2024, we have celebrated 5,000 cases.

Celebrate 5,000 cases

Da Vinci Robot

หุ่นยนต์ Da Vinci : ผู้ช่วยผ่าตัดยุคใหม่
ด้วยเทคโนโลยีล้ำสมัย ผ่าตัดแม่นยำ พื้นตัวเร็ว

ลองนึกภาพว่าคุณกำลังผ่าตัด แต่แทนที่ศัลยแพทย์
จะใช้มือ เขากลับใช้เครื่องมือจิ๋วที่ควบคุมด้วยแขนกล
ผ่านภาพ 3 มิติความละเอียดสูง นี่ไม่ใช่เรื่องที่เกิดขึ้น
ในภาพยนตร์ แต่เทคโนโลยีล้ำสมัยนี้
คือ " หุ่นยนต์ Da Vinci " อนาคตของการผ่าตัดที่
โรงพยาบาลศิริราช นำมาใช้เป็นแห่งแรกในไทย

Da Vinci ประกอบด้วย 3 ระบบหลักๆ คือ

- ชุดควบคุม : ศัลยแพทย์นั่งควบคุมการเคลื่อนไหวของแขนกลผ่านจอภาพ 3 มิติ
- ตัวหุ่นยนต์ : มีแขนกล 4 แขน 1 แขนถือกล้อง 3 แขนถือเครื่องมือผ่าตัด
- ชุดควบคุมภาพ : เชื่อมต่อระบบภาพ 3 มิติ ให้ศัลยแพทย์เห็นภาพอวัยวะภายในผู้ป่วย

ข้อดีของการผ่าตัดด้วย Da Vinci มีมากมาย เช่น

- แม่นยำ ปลอดภัย : แขนกลทำงานแม่นยำกว่ามือมนุษย์
- ผลเสีย : เจ็บน้อย พื้นตัวเร็ว นอนโรงพยาบาลน้อย
- เข้าถึงจุดที่ยาก : ผ่าตัดได้หลายประเภท

ทั้งยังครอบคลุมการผ่าตัดที่หลากหลาย เช่น

- ผ่าตัดส่องกล้องระบบทางเดินอาหาร
- ผ่าตัดส่องกล้องระบบทางเดินปัสสาวะ
- ผ่าตัดส่องกล้องระบบหัวใจ
- ผ่าตัดส่องกล้องต่อมไทรอยด์
- ผ่าตัดส่องกล้องมะเร็งปอด

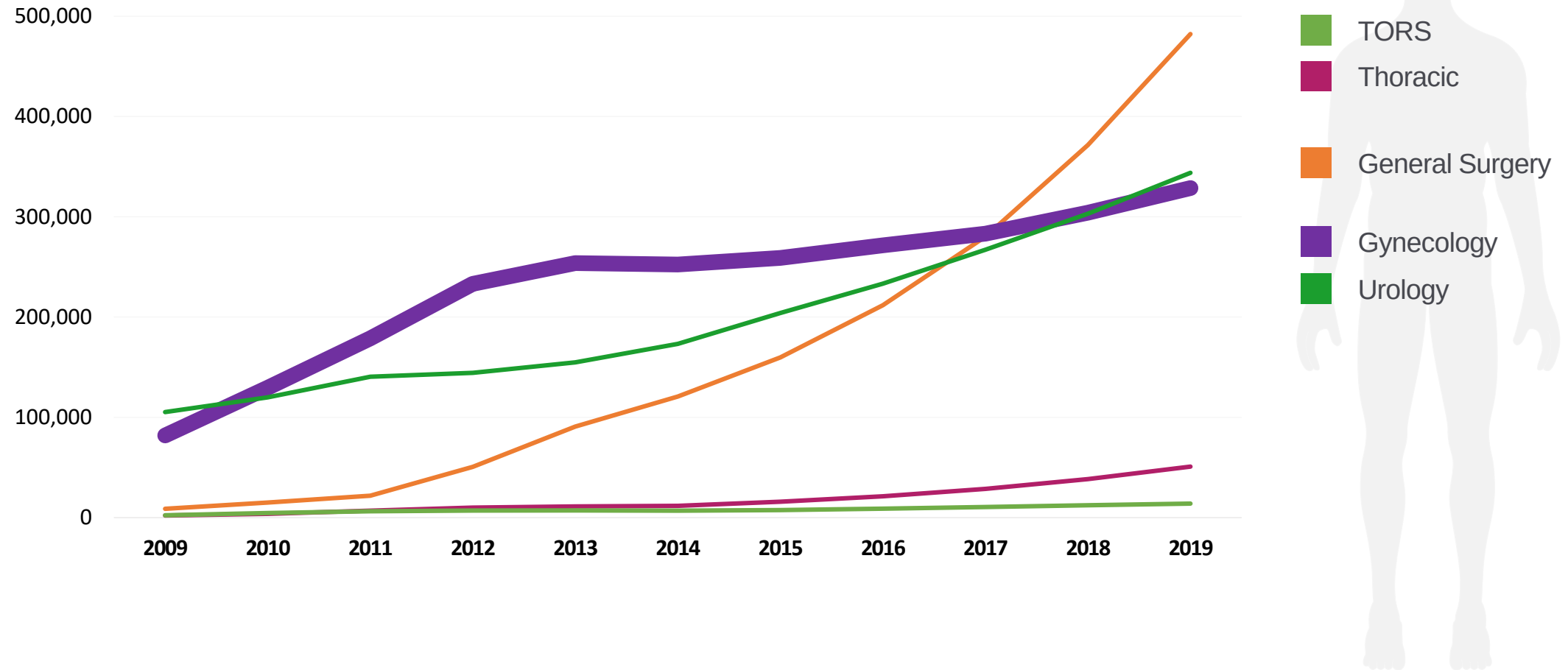
ศิริราช ศูนย์กลางการผ่าตัดด้วยหุ่นยนต์

- โรงพยาบาลศิริราชเป็นผู้นำด้านการผ่าตัดด้วยหุ่นยนต์ Da Vinci มากกว่า 10 ปี ปัจจุบันเป็นโรงพยาบาลเดียวในไทยที่มีหุ่นยนต์ Da Vinci รุ่นล่าสุดถึง 2 ตัว ควบคุมโดยศัลยแพทย์ผู้เชี่ยวชาญของศิริราชพร้อมทีมวิสัญญีแพทย์ พยาบาล และทีมสหสาขาวิชาชีพที่ทำงานร่วมกันอย่างลงตัว
- Da Vinci รุ่นล่าสุด พัฒนาให้มีขนาดเล็ก ยืดหยุ่น ควบคุมง่าย ช่วยให้ศัลยแพทย์เข้าถึงจุดที่ยากได้ขึ้น ผ่าตัดได้แม่นยำและปลอดภัยยิ่งขึ้น
- นับตั้งแต่มีหุ่นยนต์ตัวแรกในปี 2550 จนถึงปัจจุบัน ศิริราชได้ทำการผ่าตัดด้วยหุ่นยนต์ช่วยเหลือไปแล้วกว่า 5,000 ราย และรายยังคงเพิ่มขึ้นที่จะพัฒนาองค์ความรู้ สนับสนุนการวิจัย ผักอบรมศัลยแพทย์รุ่นใหม่ เพื่อให้การผ่าตัดด้วยหุ่นยนต์เข้าถึงผู้ป่วยได้มากขึ้นอีกในอนาคต

July 2021...4th Gen Davinci Xi System



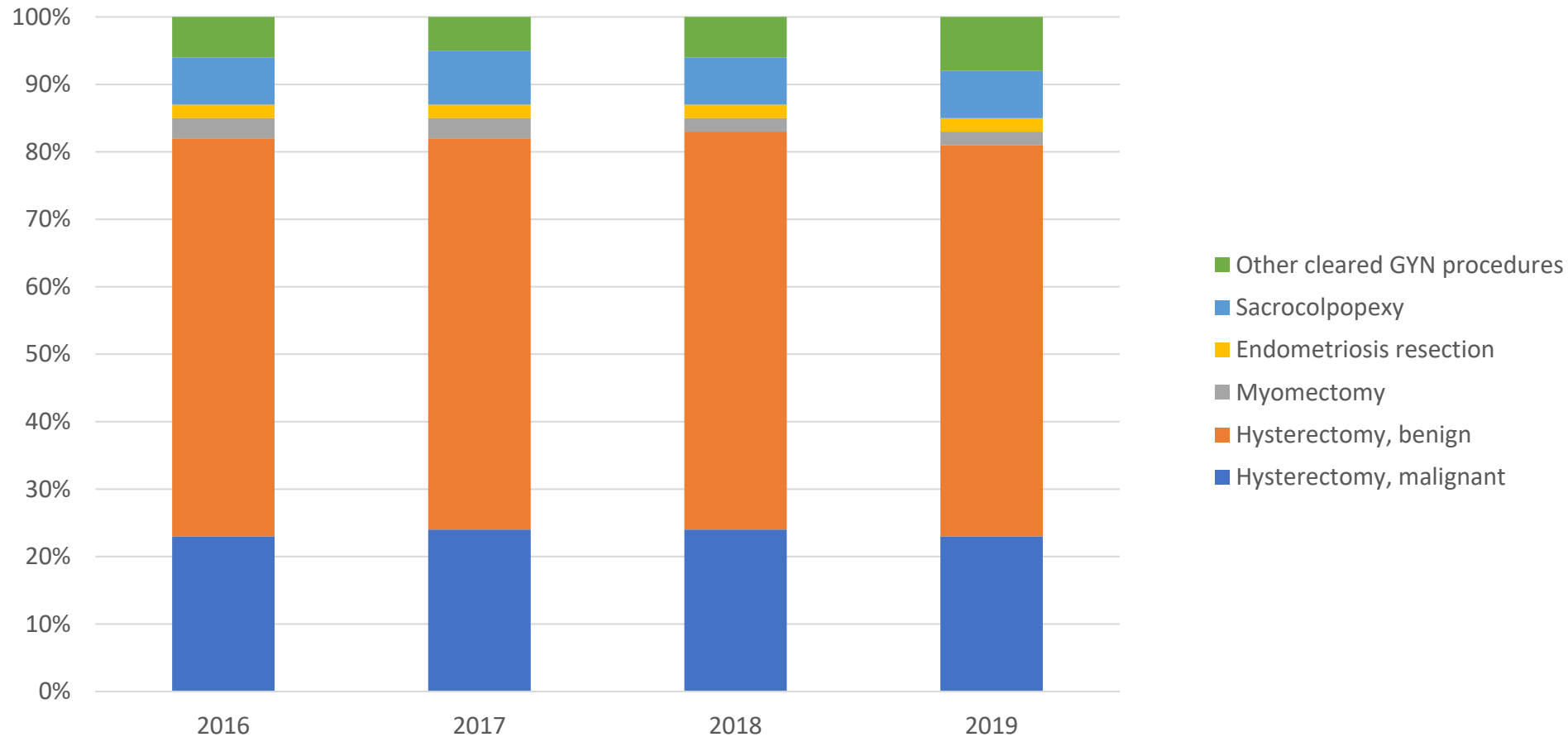
Surgeons have been performing gynecologic procedures with da Vinci systems since 2005* – a foundational legacy in da Vinci surgery**



*The da Vinci surgical system (IS2000) was cleared for use in representative gynecologic procedures in U.S. in 2005

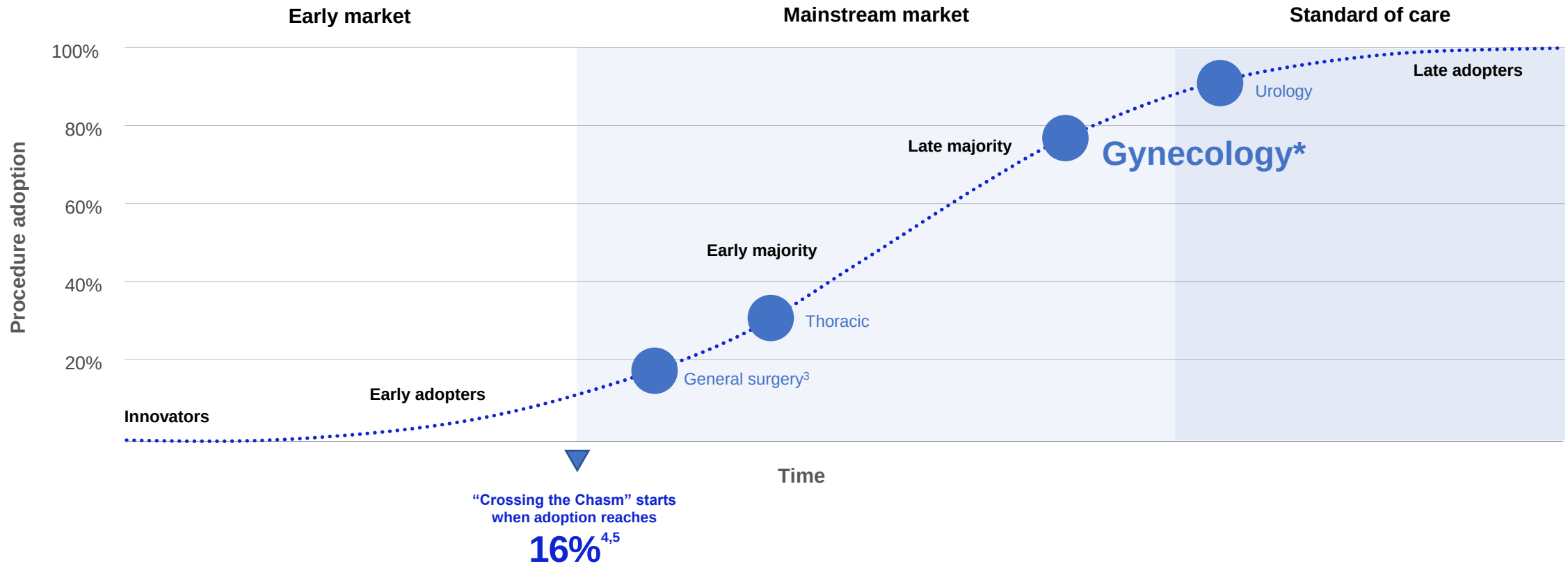
**Intuitive 2019 earnings report, global results

Surgeons are adopting da Vinci robotic-assisted surgery for appropriate patients across their Total da Vinci Practice* and complexity spectrum**



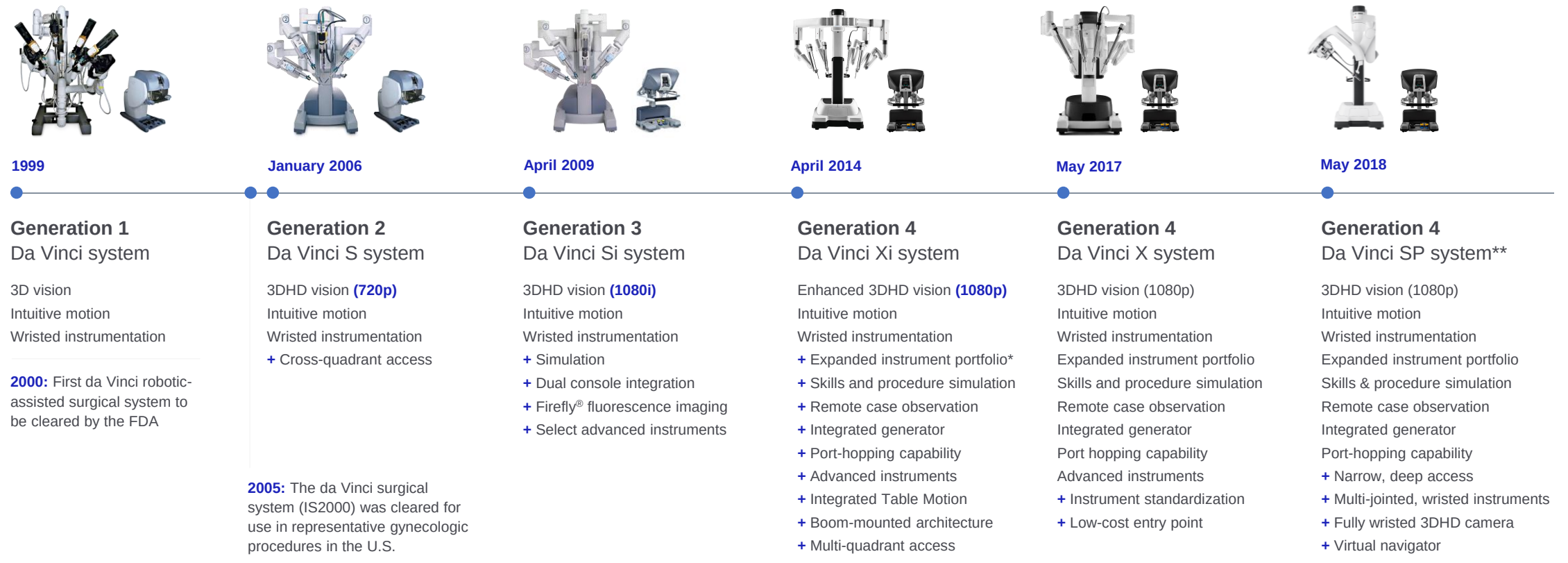
- *Total da Vinci Practice refers to the transferable value of da Vinci robotic-assisted surgery across procedures in a surgeon's minimally invasive practice. It is at the surgeon's discretion to determine when a patient is a candidate for minimally invasive surgery and whether da Vinci robotic-assisted surgery is an option.
- **Intuitive internal data on file 2016–2019

U.S. da Vinci robotic-assisted surgery adoption



- ² Intuitive internal estimates, 2018. Data on file.
- ³ Hernia is the highest volume procedure within general surgery. Intuitive internal analysis. Data on file.
- ⁴ Crossing the Chasm, Geoffrey Moore
- ⁵ Accelerating Diffusion of Innovation: Maloney’s 16% Rule
- *Gynecology procedure adoption consists of mostly malignant hysterectomy. Data on file.

20 years of advanced robotic-assisted surgery learning and development



*Compared to the da Vinci Si system.

**The safety and effectiveness of this device for use in the performance of general laparoscopic surgery procedures have not been established. This device is only intended to be used for single-port urological procedures and for transoral otolaryngology surgical procedures in the oropharynx for benign tumors and malignant tumors classified as T1 and T2 with the da Vinci EndoWrist SP Instruments and the da Vinci SP surgical system (SP1098).

The da Vinci Xi system provides a multi-faceted solution

FOURTH-GENERATION: MULTI-FACETED SOLUTIONS



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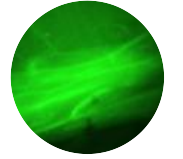
Full 30, 45, 60mm
stapling portfolio



Simulation



E-100
Generator



Sensitive
Firefly® (Xi)



Core Instruments



Integrated Energy



Vessel
Sealer Extend



Scope Flip
and Port Hopping



Integrated
Table Motion (Xi)



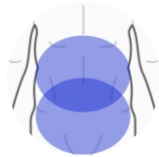
Analytics and
Advanced Software



SynchroSeal™



Endoscope Plus



Multi Quadrant
Access (Xi)



IRIS Imaging



Force Bipolar
w/ DualGrip



Handheld
Camera



The da Vinci Xi offers clinical value for abdominal surgery

1. Access and exposure



Multi-quadrant access



Integrated Table Motion



Controlled visualization
Scope flip/camera hopping



Endoscope Plus
Magnified 3D HD image



Handheld camera
Optical first entry*

2. Vascular control



Vessel Sealer Extend

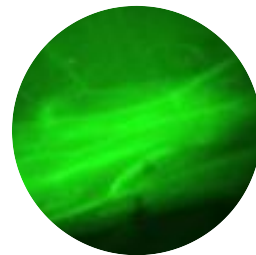


SynchroSeal



E-100 generator

3. Visualization, dissection, and defect closure



Sensitive Firefly
fluorescence imaging



Force Bipolar

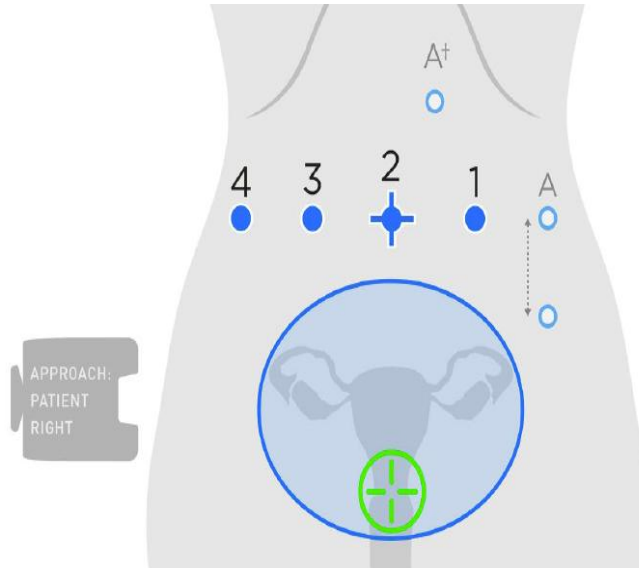


EndoWrist® Needle Driver

*Accomplish optical first entry through an 8 mm port.

Da Vinci fourth-generation technology for GYN surgery

- **Da Vinci Xi multi-quadrant access**

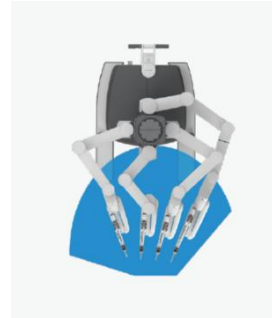


From a single dock you can access improved reach from deep in the pelvis to aortic lymph node dissection*

Workspace comparison



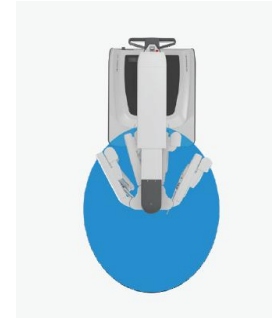
da Vinci Si



da Vinci X

1.5x greater

than da Vinci Si surgical system



da Vinci Xi

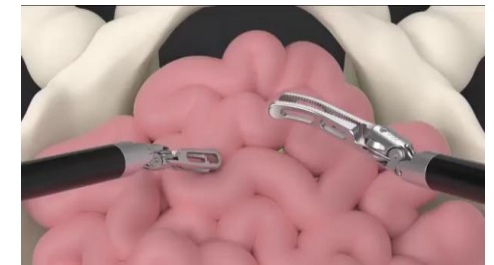
3x greater

than da Vinci Si surgical system when dual docked

Integrated Table Motion**



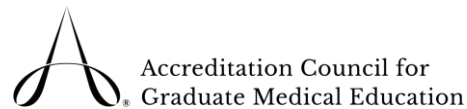
Allows surgeon to move table without stopping the case to maximize access, exposure, and reach.



* Compared to the da Vinci Si system

** The feature is only available with the da Vinci Xi system and a compatible Trumpf TruSystem™ 7000dV Table and Trumpf TruSystem™ 7000dV Remote Control by Trumpf Medical

Through 2019, there was at least one da Vinci system in each of the 50 largest residency program hospitals in the U.S.^{6 7}



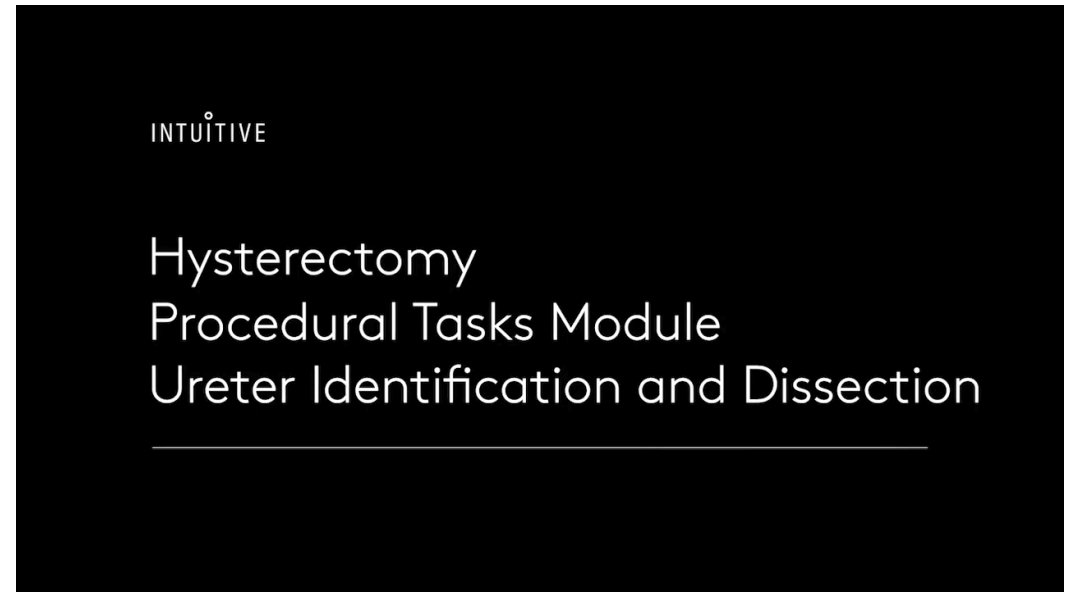
Percentage of 50 largest GYN residency program hospitals that use the da Vinci system:

92%

Gynecology (46/50 hospitals)

~350

GYN fellows trained in advanced robotics over 5 years (SGO, AUGS, AAGL)⁸



SimNow simulator

Example simulation: GYN

27% of patients

received OPEN hysterectomies in the U.S. Q4 2019?¹

⁶ List of programs provided via public data request from the Accreditation Council for Graduate Medical Education. 2018.

⁷ Information based on Intuitive internal analysis of SAP Hana data. 2018–2019.

⁸ Intuitive internal estimates, 2015–2019. Data on file.

Clinical value of da Vinci systems for GYN surgeon



Benign hysterectomy, Dr. Marc Winter



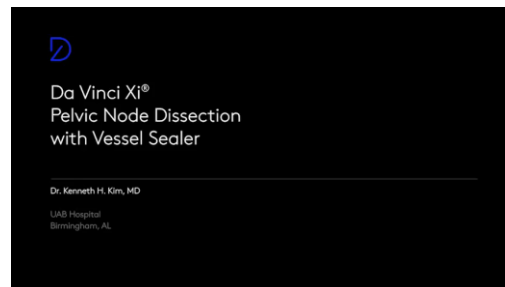
Sacrocolpopexy, Dr. Nahla Merhi



Malignant hysterectomy (Simple Total), Dr. Antonio Santillan



Myomectomy: Dr. Kathy Huang



Pelvic node dissection, Dr. Kenneth Kim



Endometriosis resection, Dr. Cindy Mosbrucker

Da Vinci for gynecologic procedures

Hysterectomy, malignant

Lymphadenectomy

Omentectomy

Parametrectomy*

Hysterectomy, benign

Salpingectomy

Oophorectomy

Adnexectomy

Cyst removal

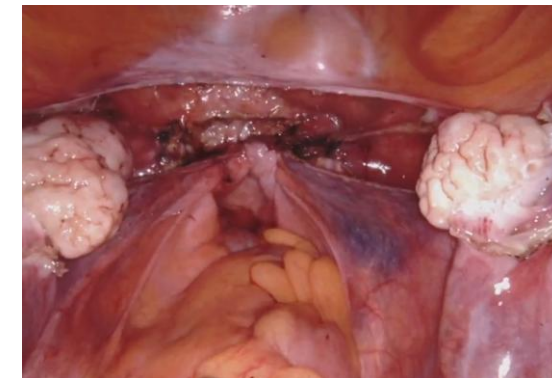
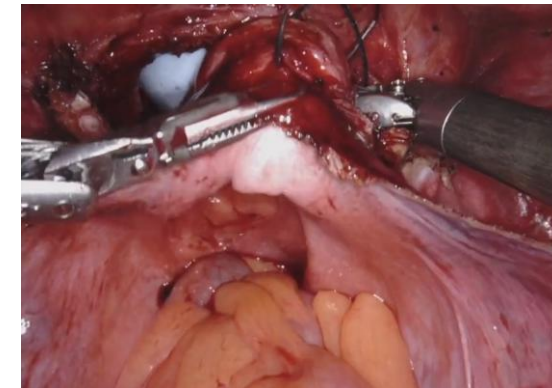
Myomectomy

Endometriosis resection

Sacrocolpopexy

* Parametrectomy is a cleared representative procedure for the da Vinci Xi system. It is not a cleared procedure for da Vinci S and Si systems.

Single-surgeon experience demonstrating EndoWrist instruments with the da Vinci Xi system in multilayer closure of vaginal cuff



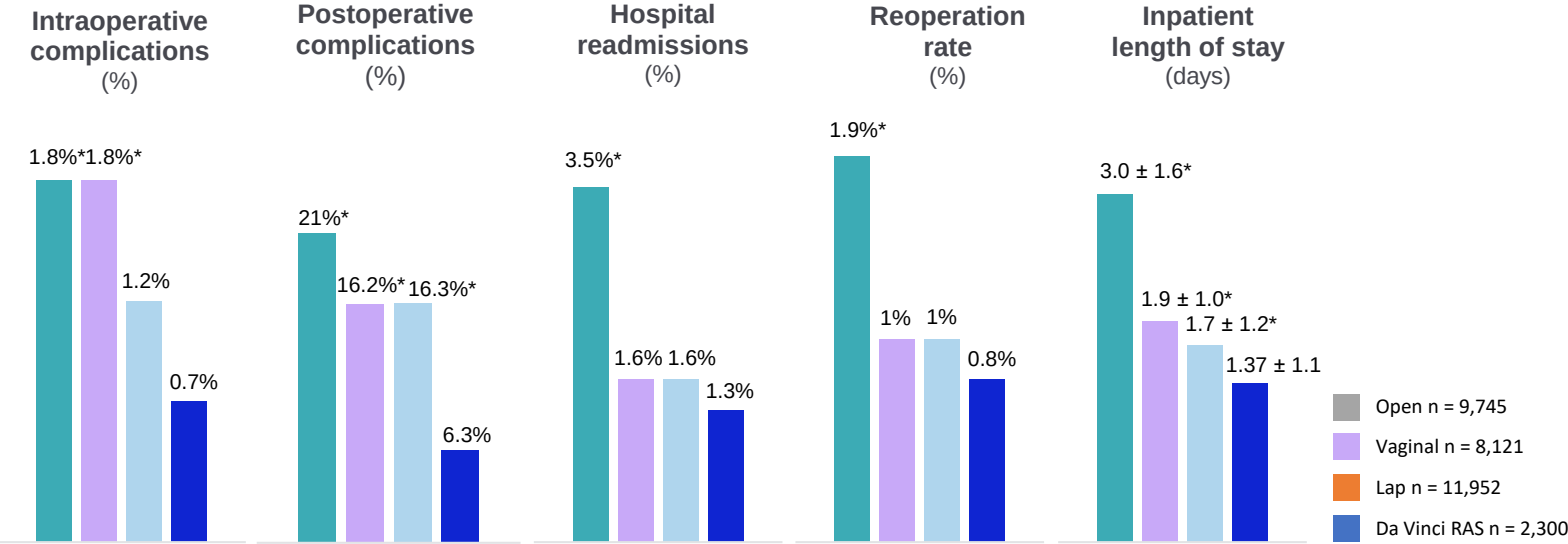
Dr. Darin Swainston
Las Vegas, NV
System training 2009

Incorporation of uterosacral ligament in vaginal cuff closure following benign hysterectomy

Multicenter analysis comparing robotic-assisted, open, lap, vaginal hysterectomies by high-volume surgeons



Dr. Peter Lim
Reno, NV
System training 2004



Analysis compares 30-day outcomes from:

- Consecutive, RAS procedures performed by high-volume surgeons (≥ 60 prior procedures) from 1/1/12 to 9/30/13
- da Vinci data from 9 centers compared with other cohorts from the Premier database
- Two-sided p < 0.05 was considered significant

Conclusion

When performed by gynecologic surgeons with relevant high-volume experience, robotic-assisted provided improved outcomes compared with abdominal, vaginal, and laparoscopic hysterectomies.

Study information

Choosing Wisely
An initiative of the ABIM Foundation

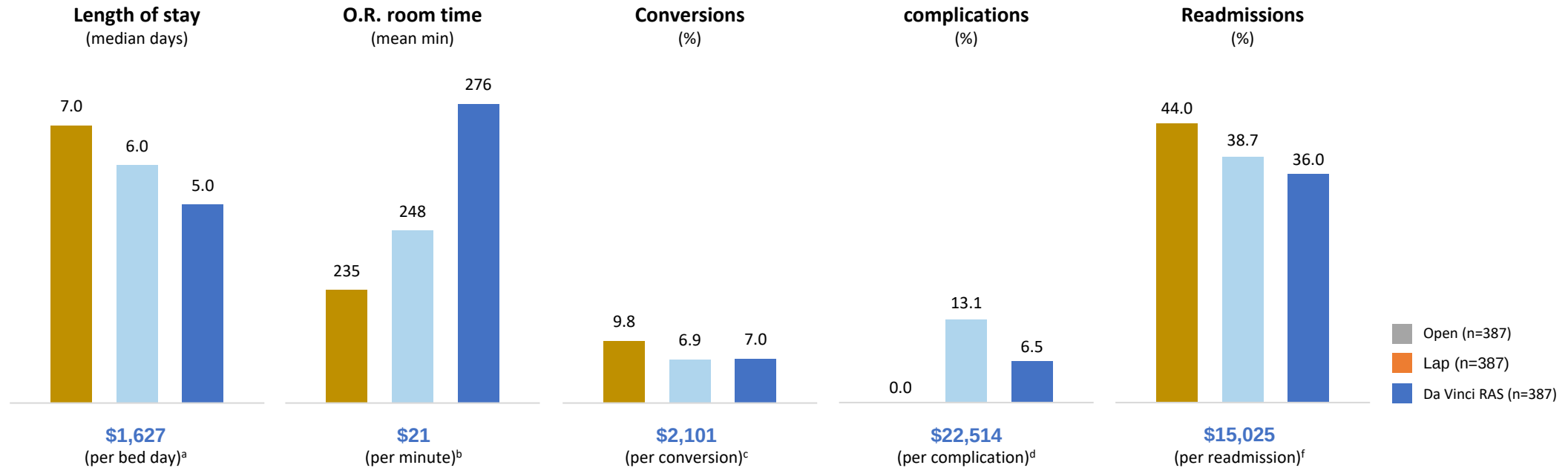
AGL
ADVANCED GYN LAPAROSCOPIC SURGERY

Five Things Patients and Providers Should Question

Do not perform a laparotomy for the management of non-malignant disease when surgical management is indicated and a vaginal, laparoscopic or robotic-assisted approach is feasible and appropriate.

Selection of an endoscopic approach should be tailored to patient selection, surgeon ability, and equipment ability. The surgeon should take into consideration how the procedure may be performed most effectively with the fewest complications.

Clinical outcomes and estimated cost avoidance



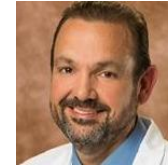
Estimated cost avoidance per procedure
\$14,645 vs. open
\$3,492 vs. lap

Estimated total cost avoidance
\$732,230 vs. open
\$174,598 vs. lap

Example Directional QTI from the the QTI Volume Estimator Tool (rev 07/2020)

Potential value in malignant procedures

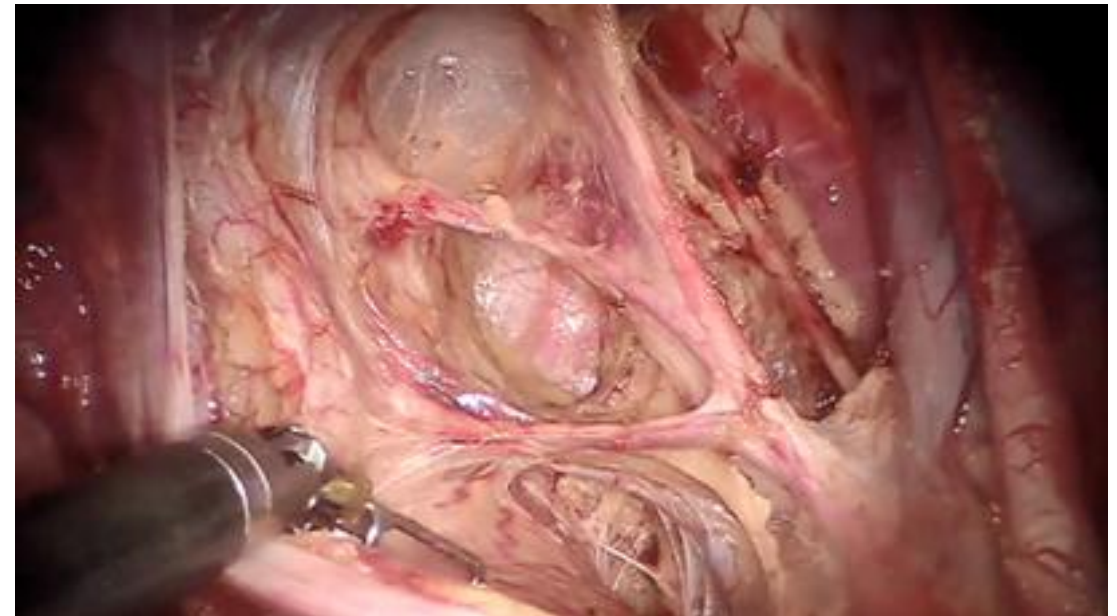
- Your da Vinci system seamlessly adjusts hand-to-instrument ratios, and tremor reduction further enhances fingertip control.
- Steady 3DHD camera provides consistent clear view of the surgical field



Dr. Ricardo Estape
Miami, FL
System training 2006



Video courtesy of Ricardo Estape, MD



Dissection of iliac vessels and obturator space

Individuals' outcomes may depend on a number of factors, including but not limited to patient characteristics, disease characteristics, and/or surgeon experience.

Study shows that hysterectomy for endometrial cancer with robotic-assisted surgery may be a better option

Outcomes*	OH	LH
Overall complications	+	✓
Intra-operative complications	+	+
Post-operative complications	+	✓
Readmissions	+	Not reported
Vaginal cuff dehiscence	–	Not reported
Transfusion	+	✓
Estimated blood loss	+	+
Length of stay	+	+
Conversion to laparotomy	N/A	+
Operating time	–	✓
Total lymph nodes counted	✓	✓
Death within 30 days	✓	✓ **

*Additional outcomes comparison disclosures located in the study information slides

** Only one study reported no death

+ Da Vinci RAS advantage over open/lap
 ✓ Da Vinci RAS comparable to open/Lap
 – Unfavorable for da Vinci RAS

Study characteristics	RH vs. OH		RH vs. LH	
No. of studies	24		24	
Total no. of patients	3,511		3,755	
Patient demographics	RH (n=1,528)	OH (n=1,983)	RH (n=1,773)	LH (n=1,982)
Age range (years)	51-78	50-77	54-67	51-69
Mean BMI range (kg/m ²)	23-51	25-54	23-51	25-48

Conclusions

- In these clinical studies, RH was associated with comparable or better clinical outcomes compared to OH and LH for patients with endometrial cancer.
- **RH was associated with:**
 - **Fewer overall complications, shorter length of stay (LOS), lower estimated blood loss (EBL), fewer transfusions, and fewer readmissions compared to OH**
 - **Shorter LOS, less EBL, and fewer intraoperative complications compared to LH**

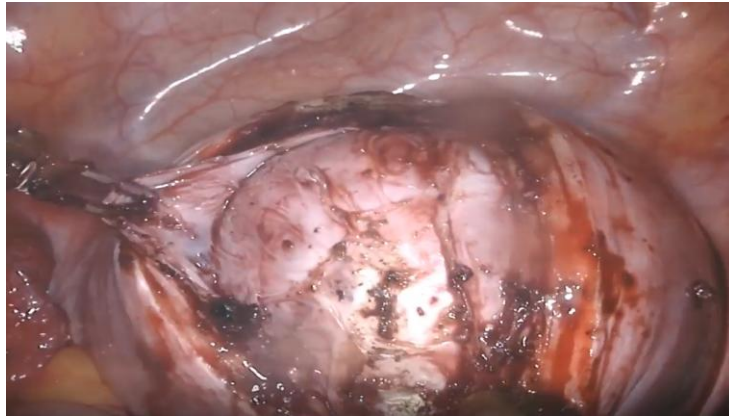
This publication evaluates the surgical safety and clinical effectiveness of robot-assisted hysterectomy (RH) compared to open hysterectomy (OH) and laparoscopic hysterectomy (LH) for endometrial cancer. This publication reports the results of a systematic review of 37 studies, which included the largest number of cases compared to those reported in previous studies

²⁶ D.A. Park, D.H. Lee, S.W. Kim, S.H. Lee, Comparative safety and effectiveness of robot-assisted laparoscopic hysterectomy versus conventional laparoscopy and laparotomy for endometrial cancer: A systematic review and meta-analysis, EJSO (2016), doi: 10.1016/j.ejso.2016.06.400

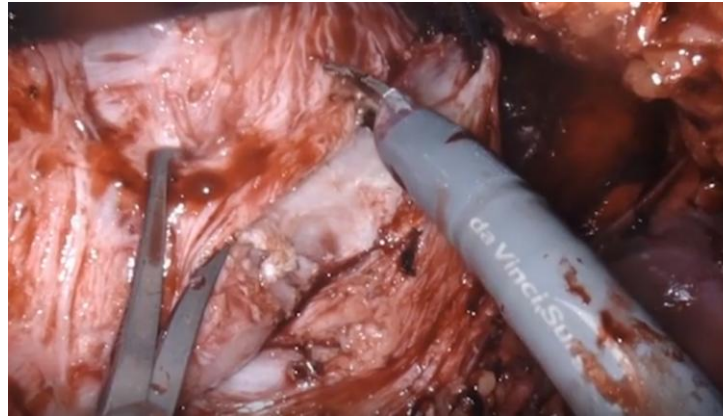
PN1079730-US RevA 01/2021

Potential value of da Vinci systems in myomectomy

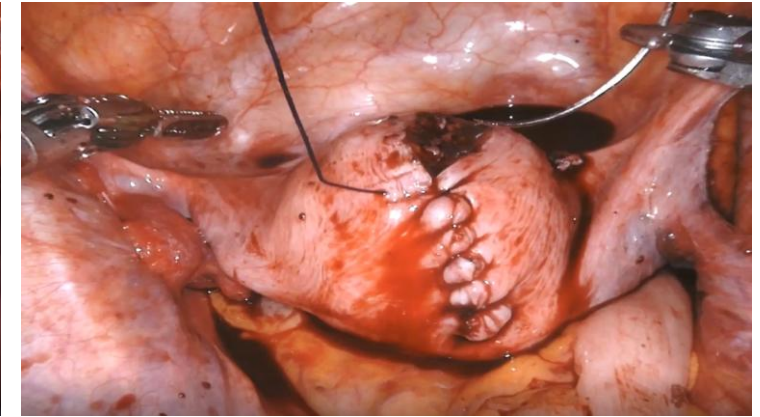
Myoma capsule dissection



Myoma removal



Suturing of hysterotomy defect(s)



3DHD vision allows you to:

- Stay oriented in the anatomy
- See tissue planes clearly
- Identify structures

Three da Vinci instrument arms:

- Have a wide range of motion to help simplify port setup and provide greater reach into the patient anatomy
- Allows control of up to two instruments with either hand, and one with the opposite hand

Wristed instrumentation facilitates:

- Fully wristed dexterity with a minimally invasive approach
- A greater range of motion than the human hand
- Suturing at multiple angles

Images courtesy of Drs. Jessica Vaught and Kathy Huang.

Individuals' outcomes may depend on a number of factors, including but not limited to patient characteristics, disease characteristics, and/or surgeon experience.

Robotic-assisted vs. laparoscopic and abdominal myomectomy for treatment of uterine fibroids: A meta-analysis

	Robotic-assisted (RALM) vs. Abdominal Myomectomy (AM)	Robotic-assisted (RALM) vs. Laparoscopic Myomectomy (LM)
Length of hospital stay	+	✓
EBL	+	+
Complications	+	+
Postoperative pain	+	Not reported

+ Da Vinci RAS advantage over open ✓ Da Vinci RAS comparable to open/lap

Meta-analysis comparing the outcomes of the three currently major surgical techniques used in patients with uterine fibroids: robotic-assisted laparoscopic myomectomy (RALM), laparoscopic myomectomy (LM), and abdominal myomectomy (AM).

Meta-analysis included 20 studies involving 2,852 patients. Most studies (14/20, 70%) were conducted in the U.S.

- Versus open: 13 studies, n = 1,825 (RALM: 773, AM: 1,079)
- Versus lap: 10 studies, n = 1,000 (RALM: 510, LM: 490)

Outcome measures included estimated blood loss (EBL), number of blood transfusions, conversions, length of hospital stay (LOHS), complications, operating time, size of the largest fibroid, fertility, bleeding, reoperation, and postoperative pain.

- Additional outcomes comparison disclosures located in the study information slides

Conclusions

RALM provides certain advantages over the classical surgical approaches regarding both the short-term and long-term outcomes.

RALM is associated with significantly fewer complications, significantly lower EBL, significantly less bleeding, and significantly fewer conversions when compared to LM.

Although operating time for RLM was significantly higher when compared to AM, the LOHS was significantly shorter with RLM.

Maximum postoperative pain score with RALM was significantly lower than AM.

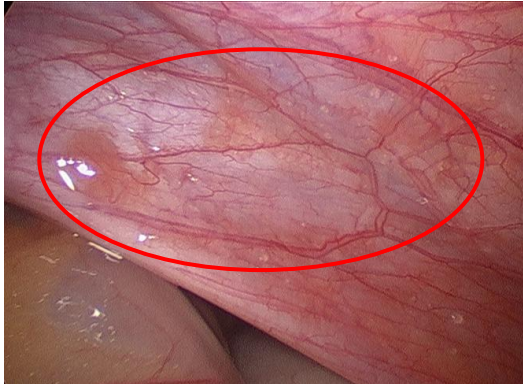
Study information

³¹ Wang, T., H. Tang, Z. Xie and S. Deng (2018). "Robotic-assisted vs. laparoscopic and abdominal myomectomy for treatment of uterine fibroids: a meta-analysis." Minimally Invasive Therapy and Allied Technologies: 1-16. DOI: 10.1080/13645706.2018.1442349.

Individuals' outcomes may depend on a number of factors, including but not limited to patient characteristics, disease characteristics, and/or surgeon experience.

Potential value of da Vinci systems in endometriosis resection

Detection of endometriosis lesions



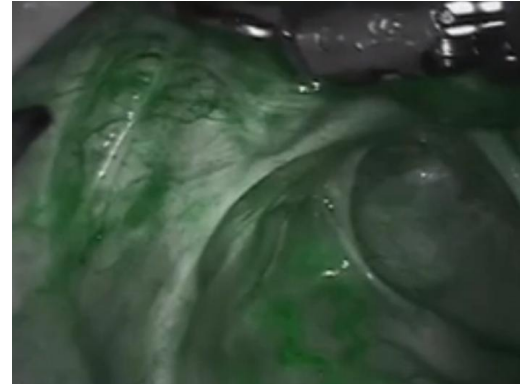
A comparative study showed that imaging with the advanced 3DHD endoscope was associated with an increased number of confirmed endometriosis lesions detected compared to 2DHD laparoscope.³²

Dissection around delicate structures



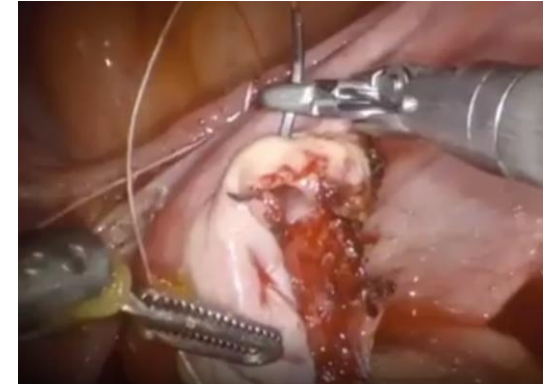
A retrospective study showed that robotic assistance greatly enhanced surgical techniques in challenging and surgically demanding procedures.³³

Identification of hypervascularity



Fluorescence imaging with Firefly technology and ICG allows real-time visual assessment of vessels, blood flow, and related tissue perfusion.

Repair following removal of endometriomas



EndoWrist technology lets you suture with a greater range of motion than the human hand, and at multiple angles.

³² Mosbrucker C, Somani A, Dulemba J. Visualization of endometriosis: comparative study of 3-dimensional robotic and 2-dimensional laparoscopic endoscopes. J Robot Surg. 2018 Mar; 12(1):59-66. doi: 10.1007/s11701-017-0686-0. <https://doi.org/10.1007/s11701-017-0686-0>

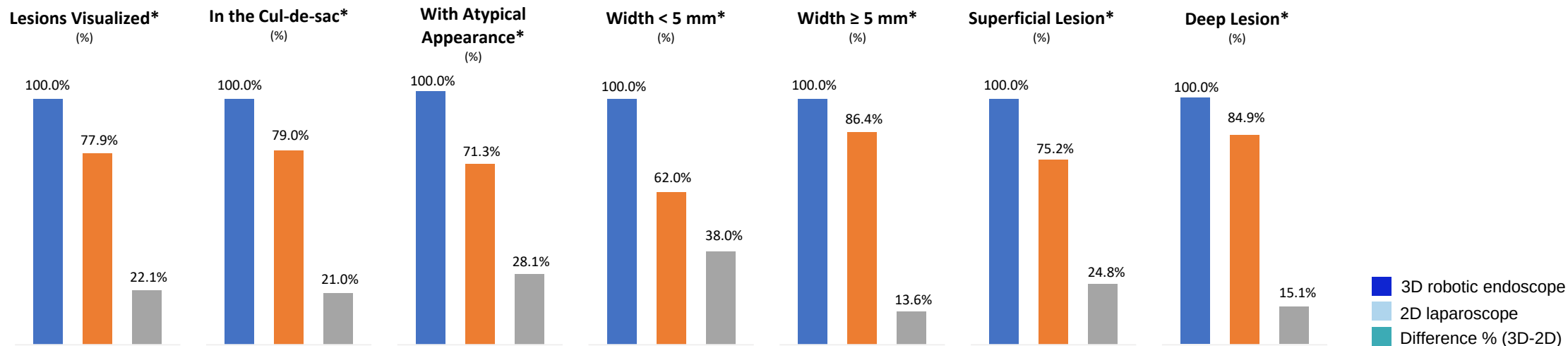
³³ Siesto G, Ieda N, Rosati R, Vitobello D. Robotic surgery for deep endometriosis: a paradigm shift. Int J Med Robotics Comput Assist Surg 2014; 10: 140-146. DOI: 10.1002

Images courtesy of Drs. Gaby Moawad, Cindy Mosbrucker, and Kenneth Levey

Additional outcomes comparison disclosures located in the background information slides.

Individuals' outcomes may depend on a number of factors, including but not limited to patient characteristics, disease characteristics, and/or surgeon experience.

Statistically significant increase in detection of confirmed endometriosis lesions with 3 robotic endoscope



*All results reported are for lesions histologically positive for endometriosis.

Objective

Compare the results of using the robotic 3DHD scope and the 2DHD laparoscope for visual detection of histologically confirmed endometriosis

- Study period: April 2011–December 2014
- Study size: 98 patients from 3 surgeons in different practices

Overview

Patients were premenopausal women aged 18 years and older who elected to undergo robotic-assisted endometriosis resection

- 598 lesions were visualized in 98 patients

Conclusions

100% of lesions confirmed as endometriosis were detected using the 3DHD scope; 77.9% were detected using the 2DHD laparoscope ($P < .001$)

Of all lesions detected, the robotic 3D scope enabled visualization of a significantly higher proportion (99.5%) compared to the 2D laparoscope (79.3%; $P < .001$)

Compared to lap visualization, robotic visualization detected more confirmed lesions in all anatomic locations and for most appearances, including cul-de-sac, atypical appearance, and width < 5 mm ($P < .001$)

Study information

³² Mosbrucker, C., A. Somani and J. Dulemba (2018). "Visualization of endometriosis: comparative study of 3-dimensional robotic and 2-dimensional laparoscopic endoscopes." J Robot Surg. DOI: 10.1007/s11701-017-0686-0

Study suggests more accurate visualization and excision of endometriosis with robotic-assisted surgery vs. laparoscopy

	Robotic-assisted (RAS)	Laparoscopy (LS)	P Value
Operative time (minutes)	77.4	72.3	.23
Patients w/confirmed endometriosis	80.0%	56.8%	< .0001
Appendectomy	23.3%	30.0%	.32
Appendix positive for endometriosis	28.6%	3.3%	.02

- Retrospective study examines the feasibility of treating patients with suspected endometriosis using robotic-assisted laparoscopic techniques.
- Operative characteristics were analyzed for last 100 standard laparoscopic (December 2004–September 2007) and the first 180 robotic-assisted (July 2007–January 2010) surgeries to treat suspected endometriosis.
- Perioperative outcomes and postoperative pain were compared by technique including operative time, blood loss, and complication rates.
- Additional outcomes comparison disclosures are located in the study information slides.

Conclusions

Demonstrates that RAS for treatment of endometriosis is feasible across all disease stages with low morbidity and operative time comparable to LS.

Rates of suspected endometriosis upon histological confirmation were significantly higher for RAS.

RAS, compared to LS, may provide more accurate visualization and, thus, excision of existing endometriosis.

Study information

³⁴ Dulemba JF, Pelzel C, Hubert HB. (2013). "Retrospective analysis of robotic-assisted vs. standard laparoscopy in the treatment of pelvic pain indicative of endometriosis." J Robotic Surg. 7.2 (2013): 163-169 DOI: 10.1007/s11701-012-0361-4.

Individuals' outcomes may depend on a number of factors, including but not limited to patient characteristics, disease characteristics, and/or surgeon experience.

Surgical robots available in Japan



Da Vinci Xi



Da Vinci SP



Da Vinci V



Saroa



Hugo

Perioperative Outcomes

- Early-stage endometrial cancer
- Robotic-assisted extrafascial~ modified radical hysterectomy + BSO*

Case (n=45)	Da Vinci (n=35)	hinotori TM (n=10)
Total operative time (minutes)	182 (147-204)	169(153-206)
Console time (minute)	142 (110-169)	132 (125-172)
Estimated blood loss (mL)	15 (5-28)	18 (7-30)
Length of stay (days)	4 (3-4)	4
Clavien-Dindo($\geq$ III) complication (n)	0	0

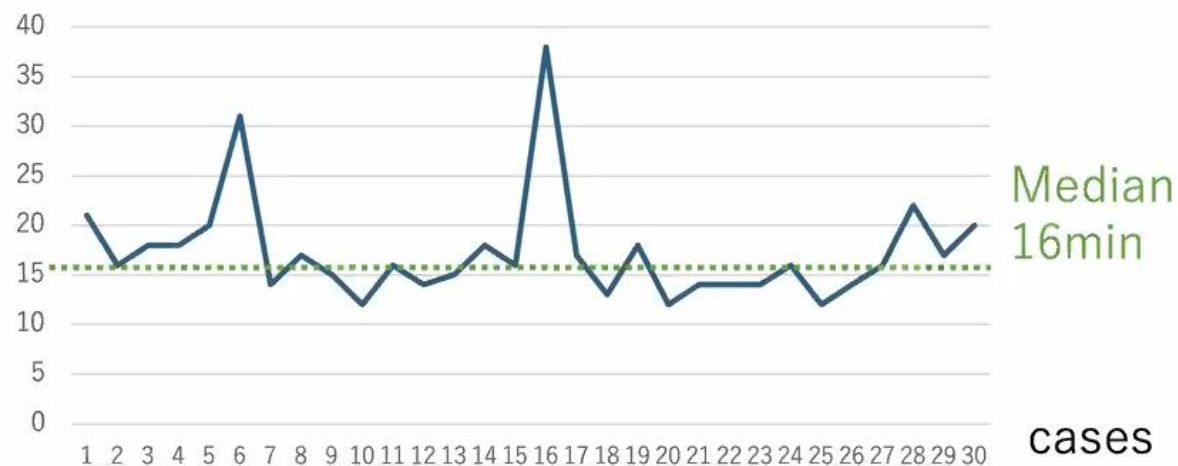
median (IQR)

*BSO; bilateral salpingo-oophorectomy

Comparison setting time of hinotori™ and Da Vinci

	hinotori	Da Vinci
cases	30	75
Start of surgery~ Roll-in (min)	7(4-26)	10(3-40)
Roll in ~ start of cockpit (min)	9(4-22)	5(2-12)
Total setting time (min)	16(12-38)	16(7-46)

(min) Total setting time median (range)



Comparison of hinotori™ and Da vinci



hinotori™

Master-slave system

Surgeon Cockpit, Operation Unit, Monitor cart

4 arms

Docking free design

16:9 screen

Made in Japan



Da Vinci Xi

Master-slave system

Surgeon console, Patient cart, Vision cart

4 arms

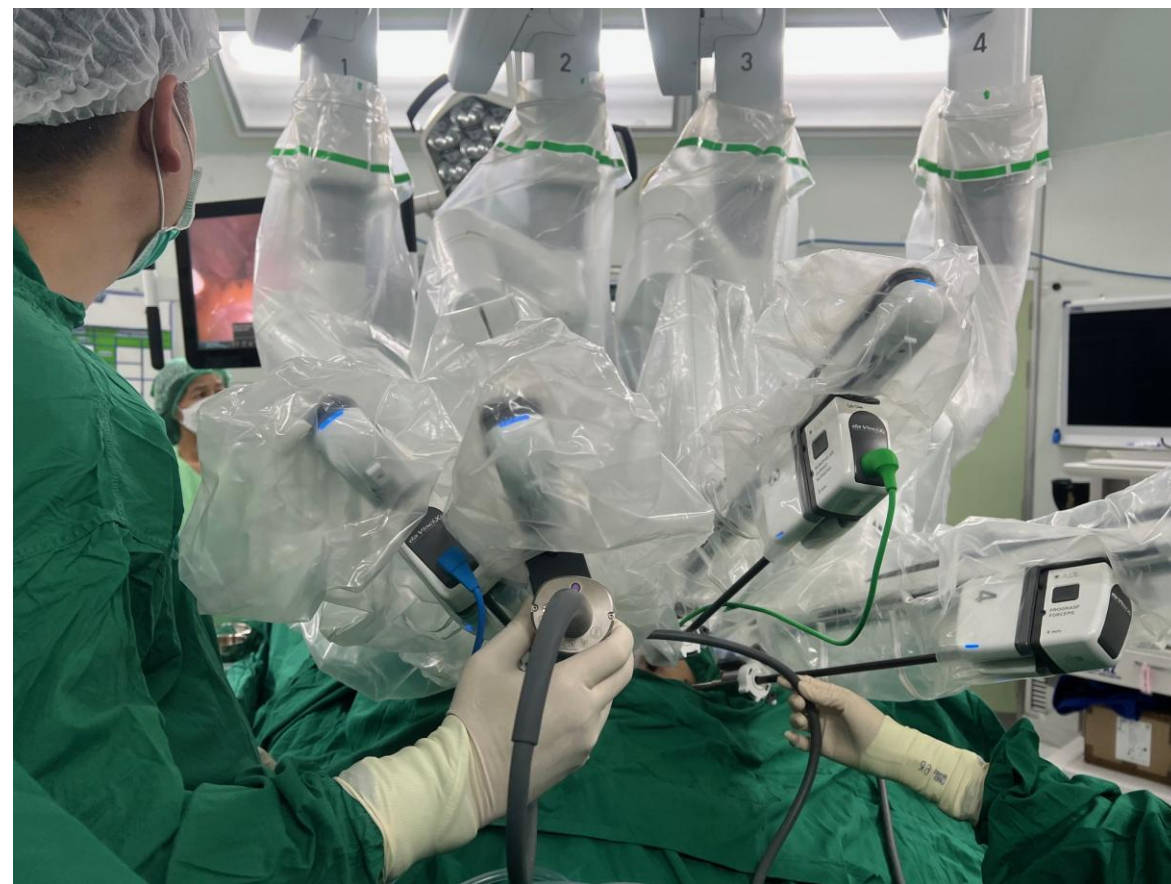
Docking arms to ports

4:3 screen

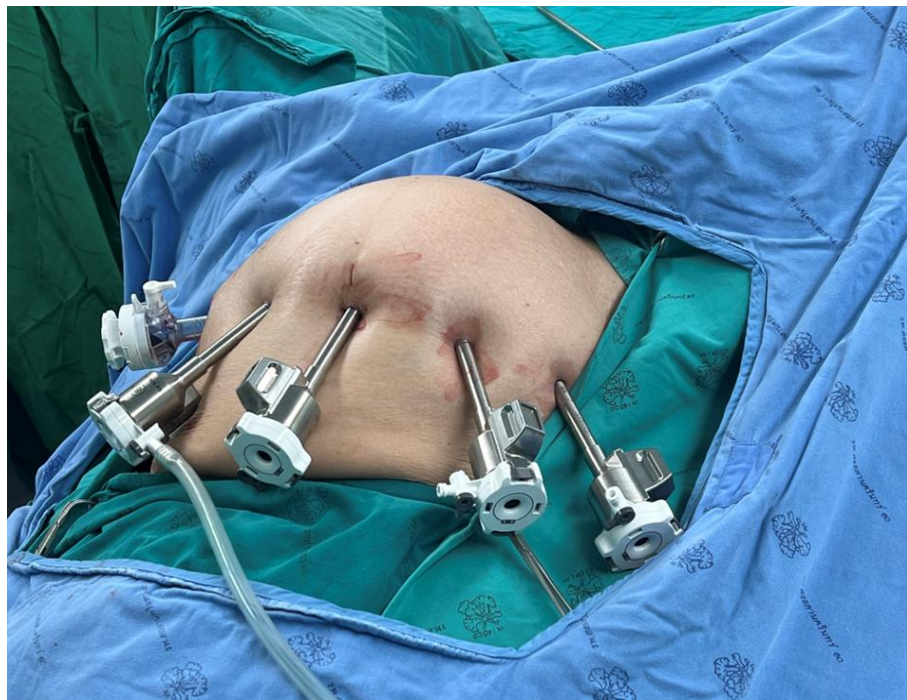
Made in United States

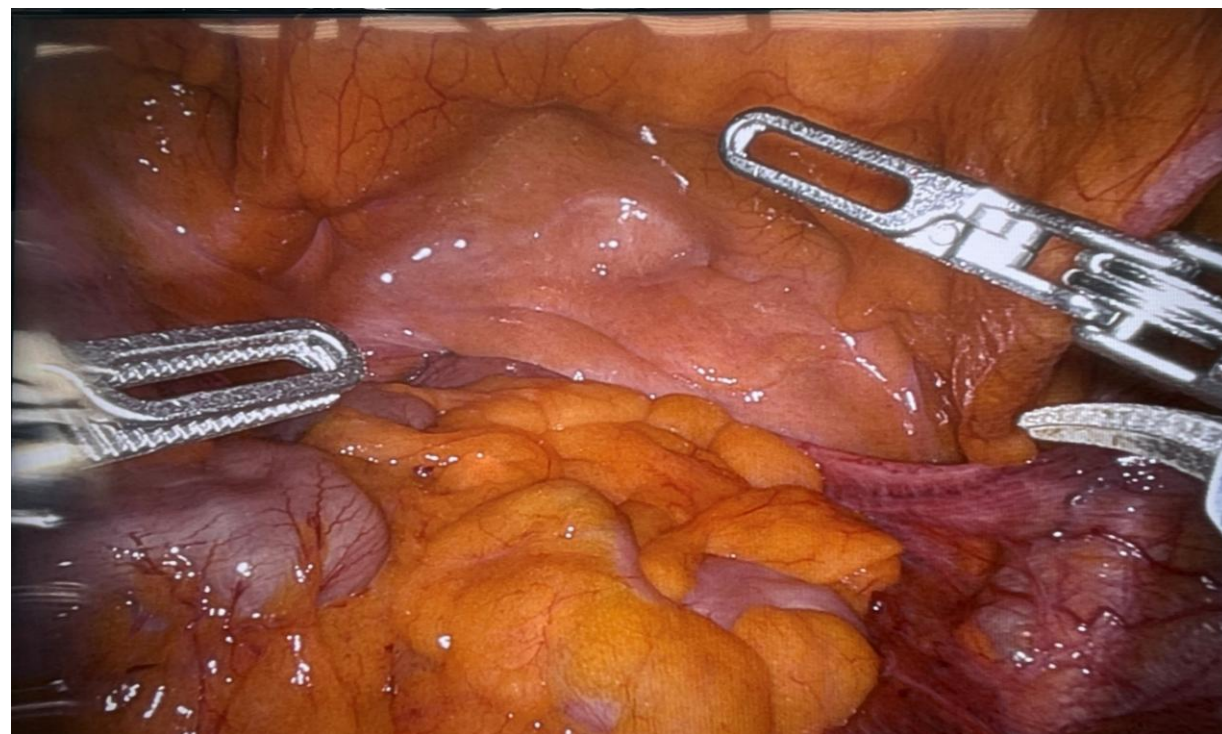
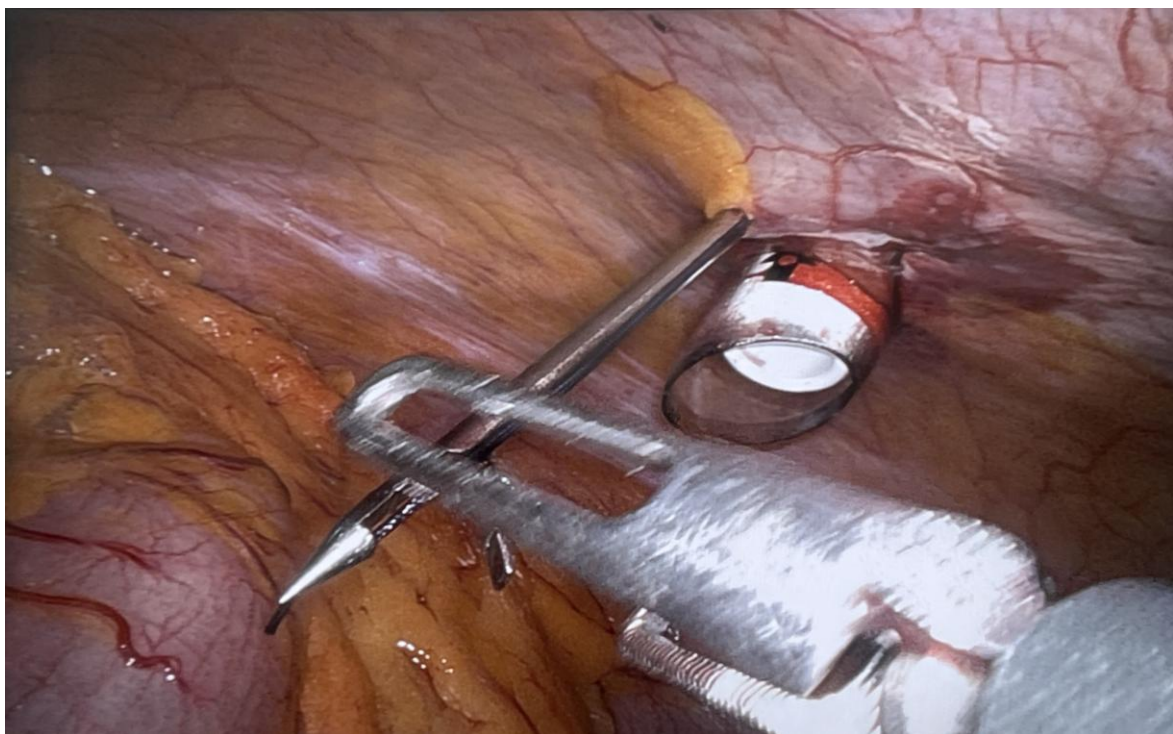


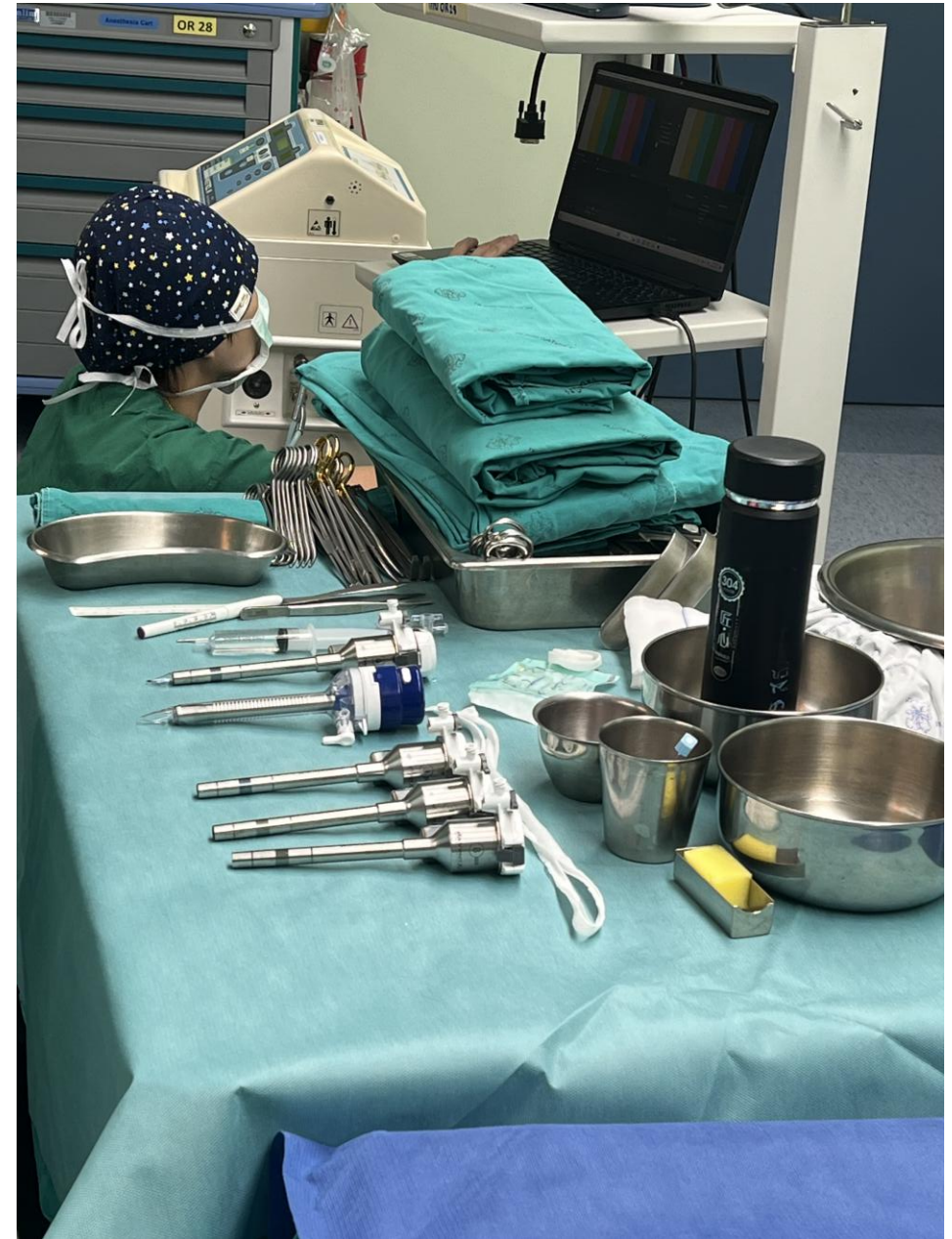




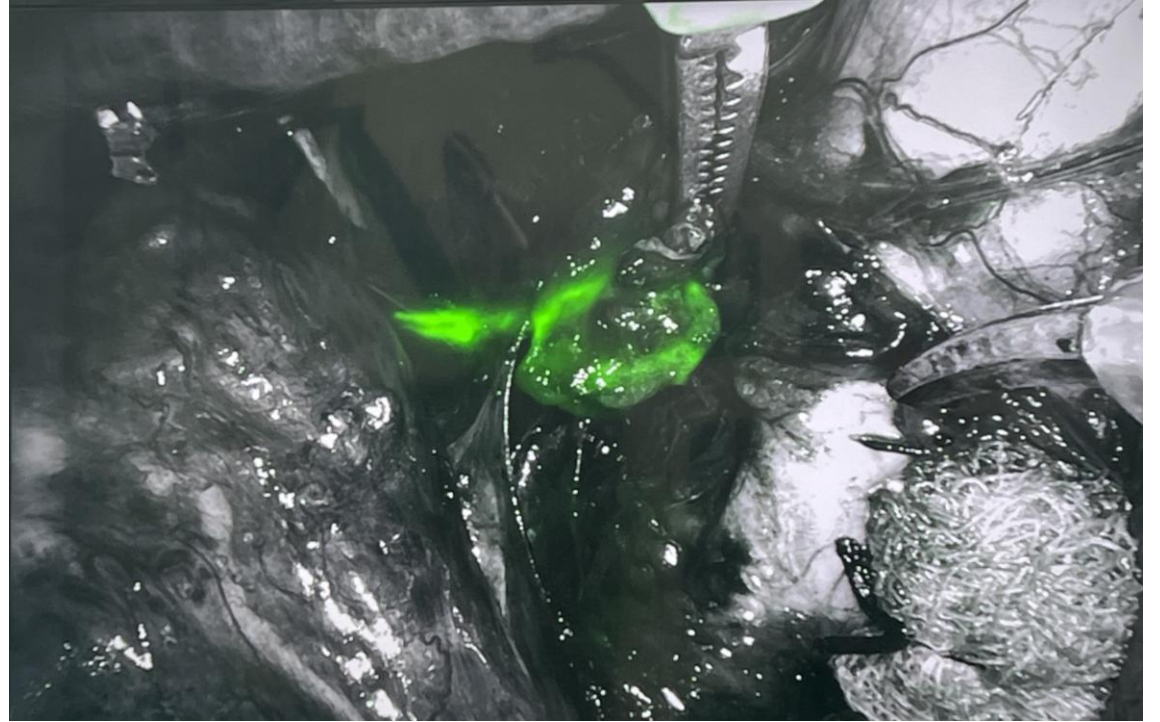


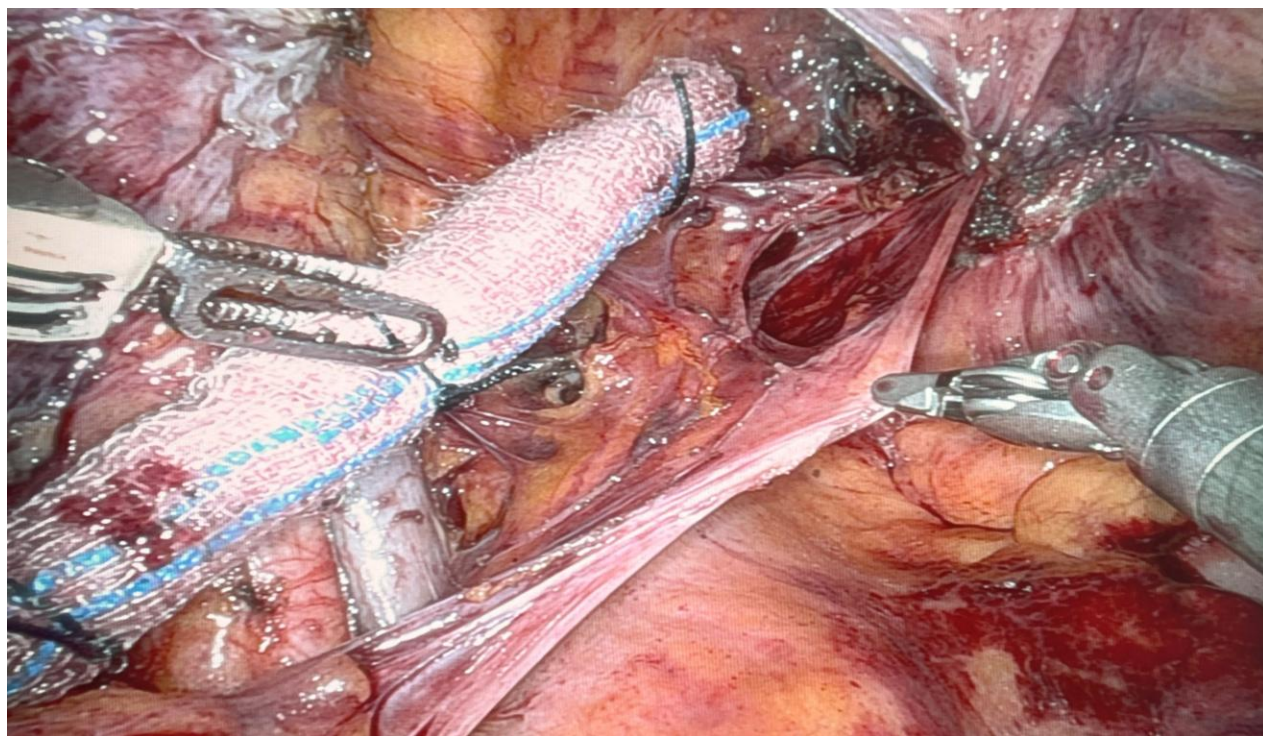
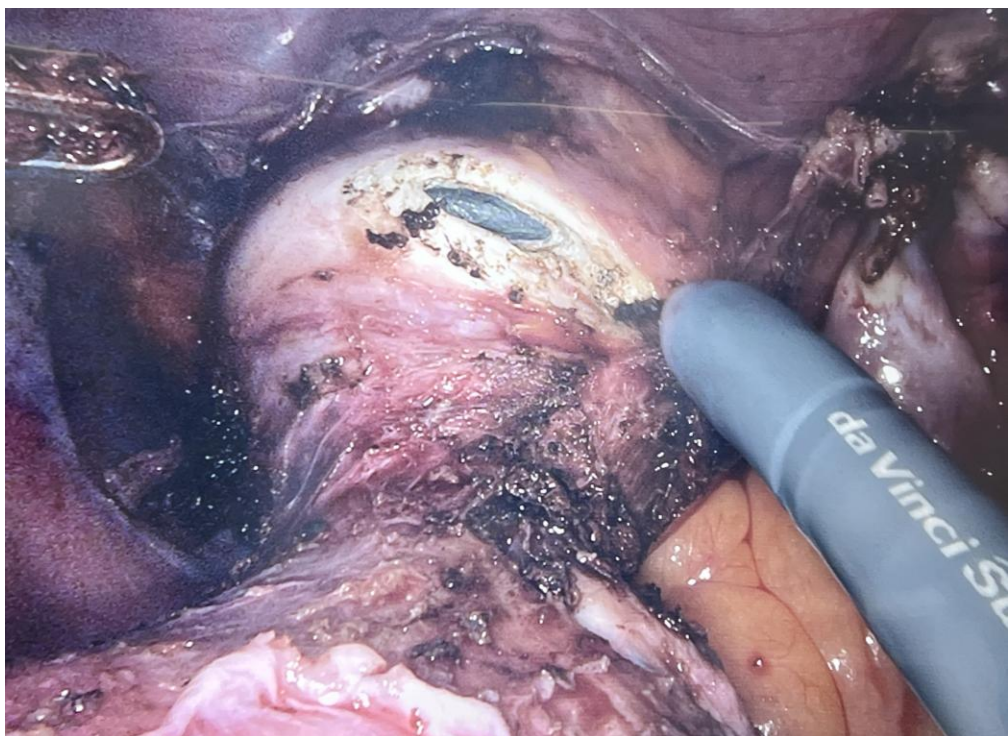














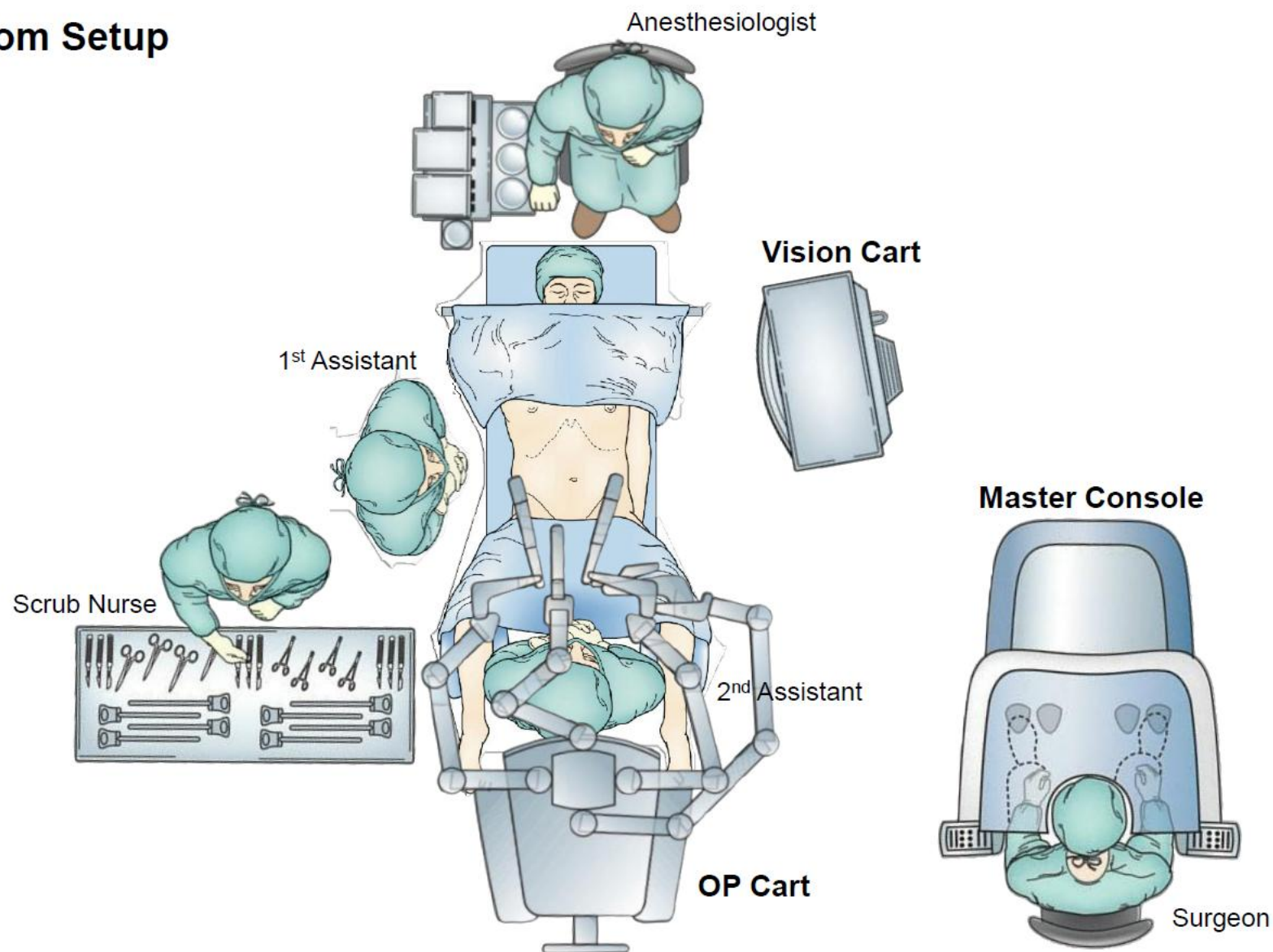
revo-1

ROBOTIC SURGICAL SYSTEM

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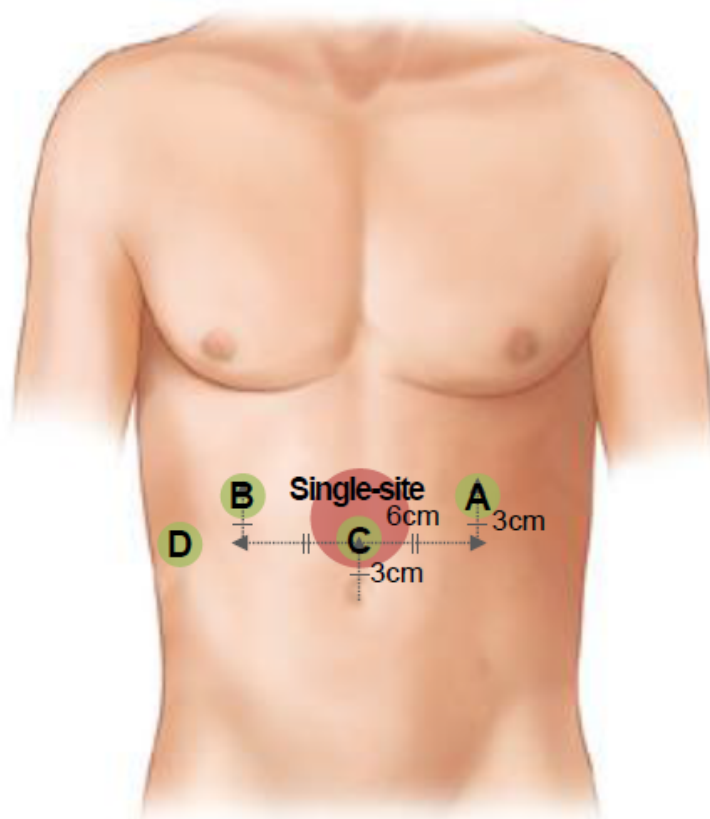
OR Room Setup



Port Placement

A. Hybrid Multi-Site Approach

A port placement strategy that integrates a single-incision laparoscopic access device with multi-site robotic trocars to enhance flexibility in robotic-assisted surgery.



① C Port (12mm) & Assist Port - Sin

- **Location:** 3cm superior to Umbilicus
- **Trocar:** Single-Port Laparoscopic Access System (SILS Port)

② A & B Ports (8mm)

- **Location:** 6cm lateral and 3cm superior to the C port

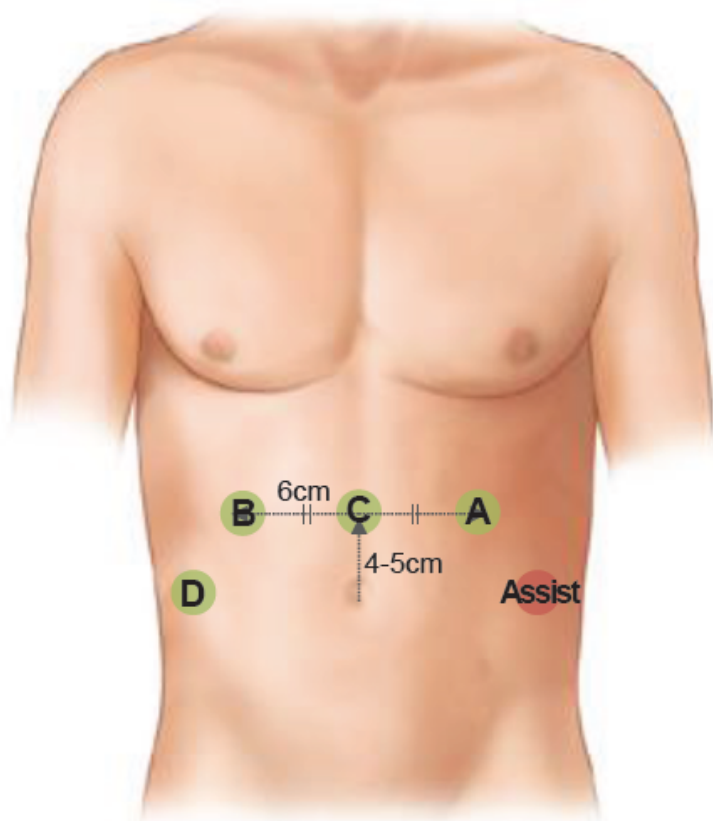
③ D Port (8mm)

- **Location:** Right lateral & inferior relative to B port
- **Distance:** Maintain 8cm between B and D ports to avoid B-D arm collision

Port Placement

B. Standard Multi-Site Robotic Approach

A conventional port placement method utilizing only multi-site robotic trocars for streamlined robotic-assisted surgery.



① C Port (12mm)

- Location: 4–5cm superior to the Umbilicus

② A & B Ports (8mm)

- Location: 6cm lateral from the C port

③ D Port (8mm)

- Location: Right lateral & inferior relative to B port, at or superior to the umbilical level
- Distance: Maintain 8cm between B and D ports to avoid B-D arm collision

④ Assist Port

- Location: Left lateral & inferior relative to A port, at or superior to the umbilical level

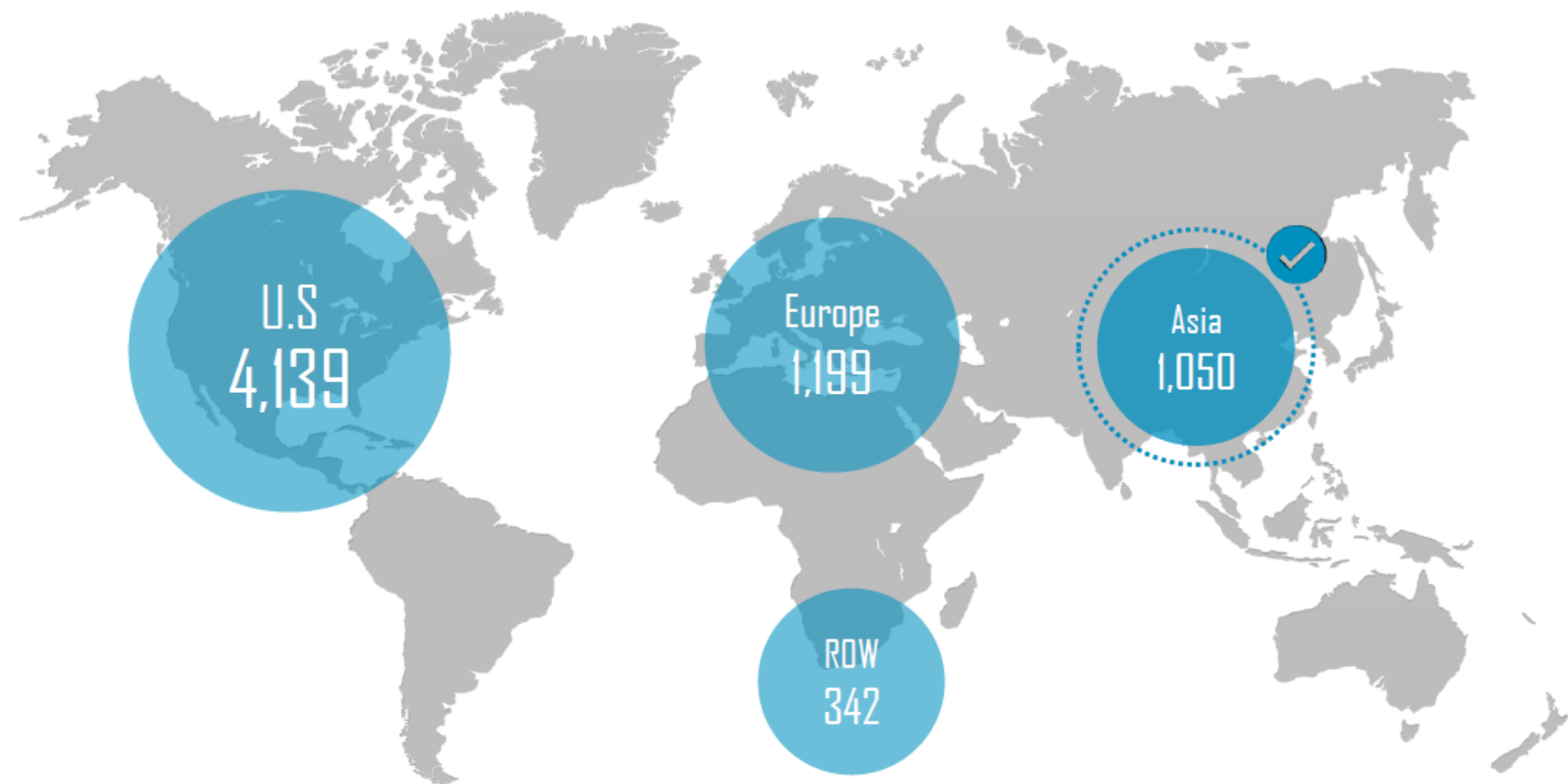
Conclusion

- Para-aortic lymph node dissection using Revo-i can be performed with **either the Hybrid Multi-Site Approach or the Conventional Multi-Site Approach.**
- The procedure can be completed with **a Single-Docking approach**, eliminating the need for Dual Docking.
- This approach **enables effective lymph node dissection from Level 1 to Level 4**, facilitating Radical Hysterectomy

2

New Era of Surgical Robot Systems

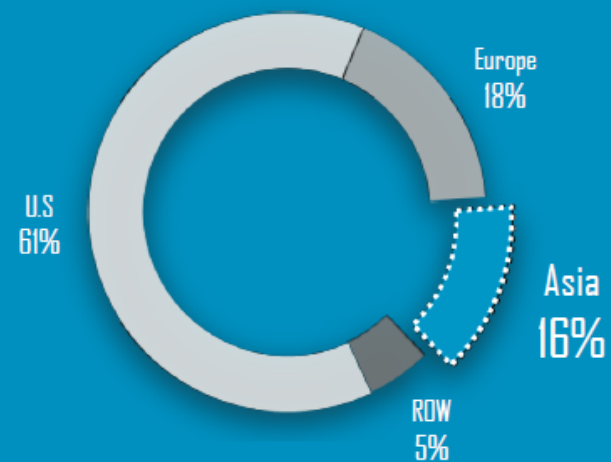
GLOBAL INSTALLED BASE (FY2021: 6,730 UNITS)



Source: Market Intelligence

revo-i

Proportion by Continent



Growth rate (FY2021 vs PY)

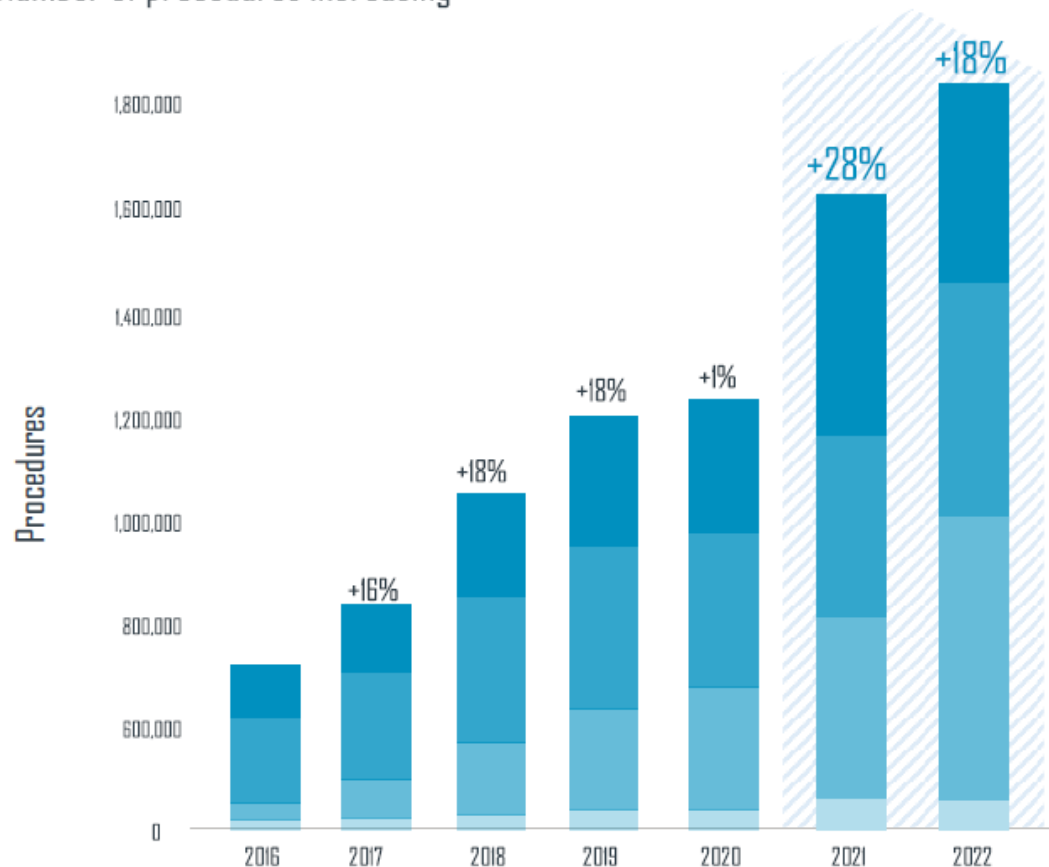
U.S.	+ 11%
Europe	+ 13%
Asia	+ 17%
Row	+ 8%

Reimbursement Approval of Japan for Laparoscopic Surgical Robot. Now more than 400 systems installed in Japan

3 New Era of Surgical Robot Systems

ROBOTIC SURGERY IS CONSISTENTLY EXPANDING

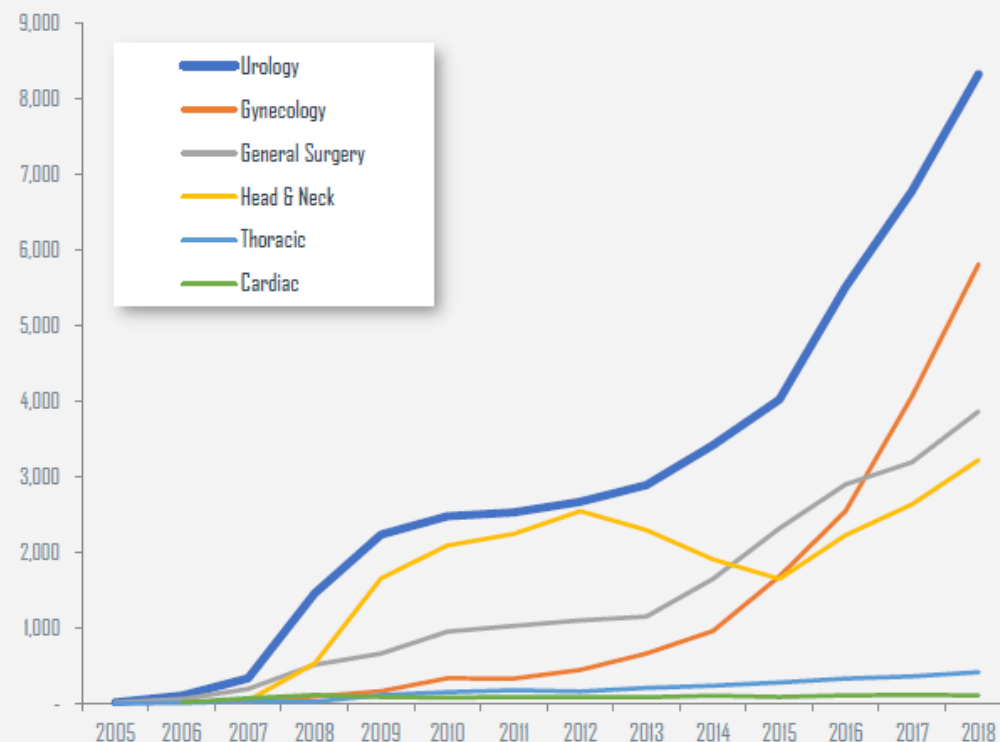
Number of procedures increasing



Source: Intuitive 2022 earning estimate

revo-i

Growth in procedure categories Korea over past 10 years



CAGR: Compound Annual Growth Rate, 연평균 복합성장률

Source: Market Intelligence



meerecompany
Revo-i (Korea)



Intuitive Surgical
Davinci Xi (U.S)



CMR Surgical
Versius (UK)



Medtronic
Hugo (U.S)



Avatera medical
Avatera (Germany)



Senhence Surgical
Asensus (U.S)



Medcaroid
Hinotori (Japan)

Optimized User Interface
for Minimizing Surgeon's Fatigue



Closed Console
Excellent User Interface

01

- Finger & Pedal vs other types
- Closed Console vs Open Console

Efficient use of space,
Ease for staffs to co-work with a surgeon



Robotic Type

02

- Standalone vs Modular

Safety for patients



Type of
Remote Center Motion

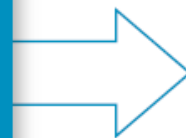
03

- Mechanical RCM vs Software RCM



User Interface

- Finger clutching
- Pedal clutching
- Switch & Buttons
- Graphic

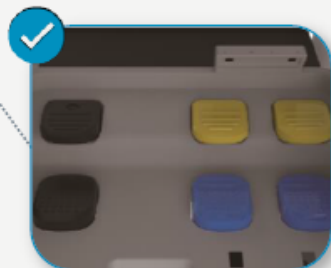


| Easy to **learn**, Easy to **use**

- ✓ Intuitive movement
 - ✓ Similar to using conventional interface
 - ✓ Minimize efforts to learn new instruction
- A more user-friendly design



Finger clutching



Pedal clutching



Grasping handle



Pedalless

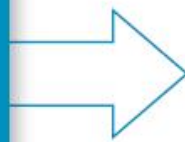


Joystick



Console Type

- Open Console
(3D Glasses)
- Closed Console



Which display type is recommendable?

→ For surgeon's **eye fatigue**



Open Console

- Small and lightweight console
- High eye fatigue can also result from 3D glasses wearing
- In comparison to closed console, 3D depth comparison analysis is required

VS

Closed Console

- Superior surgical immersion
- Low level of visual fatigue





02 Robotic Type

- Standalone
- Modular

| Keep your space and time available for setting systems

- ✓ Space for surgical field and operation room
→ "Is it actually **effective**?"



Standalone

- **Drape easily** at the designated spot
- **Does not occupy much space** to support operation



VS

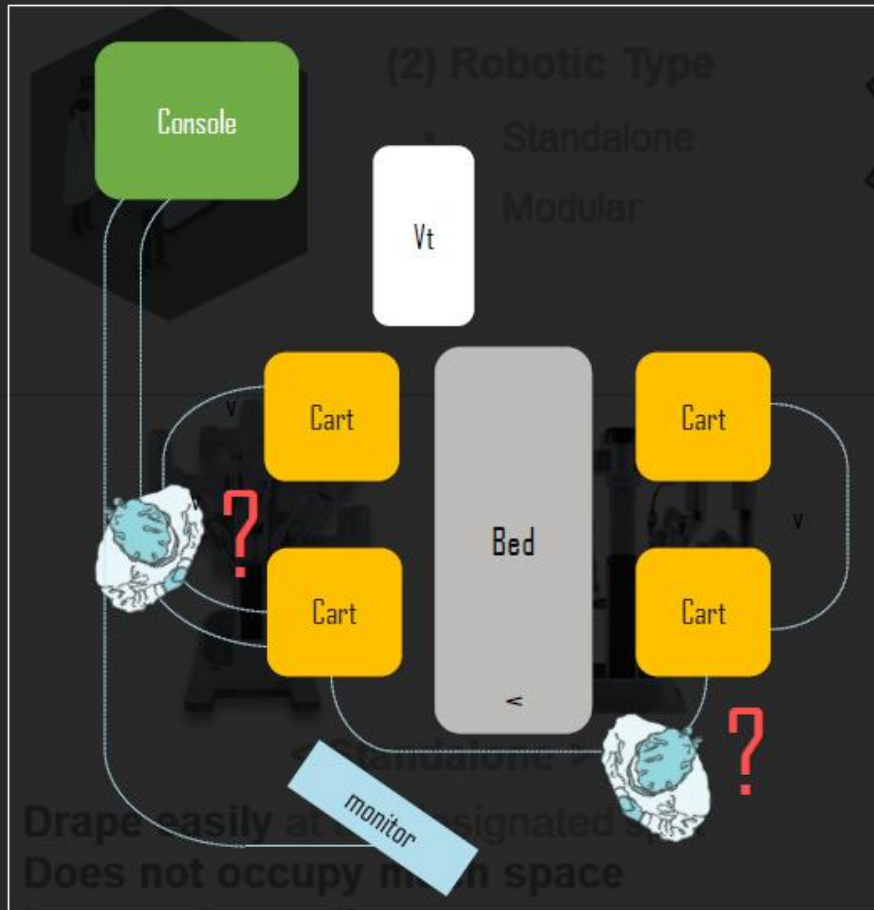
Modular

- Surgically prep and position each arm individually
- Presenting a **tight workspace**
- Different surgical procedures need for **different arm placement guide** (Distance, Location, Line connection)



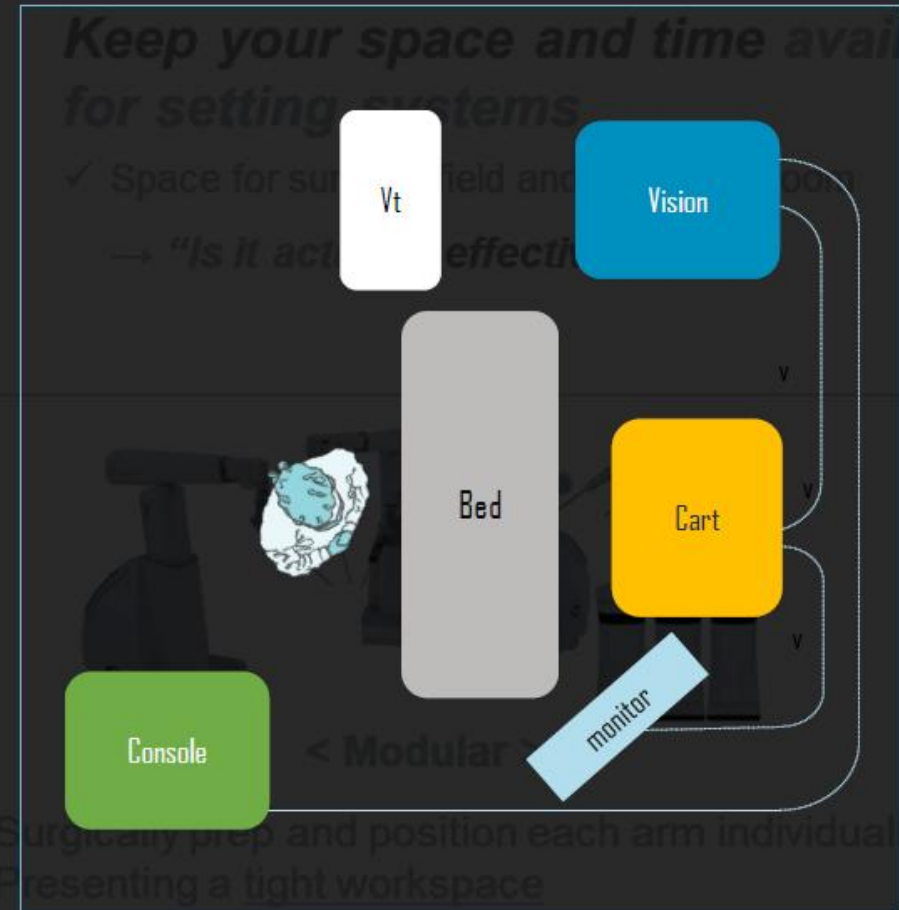
The Technological Value of Surgical Robot

Modular Type

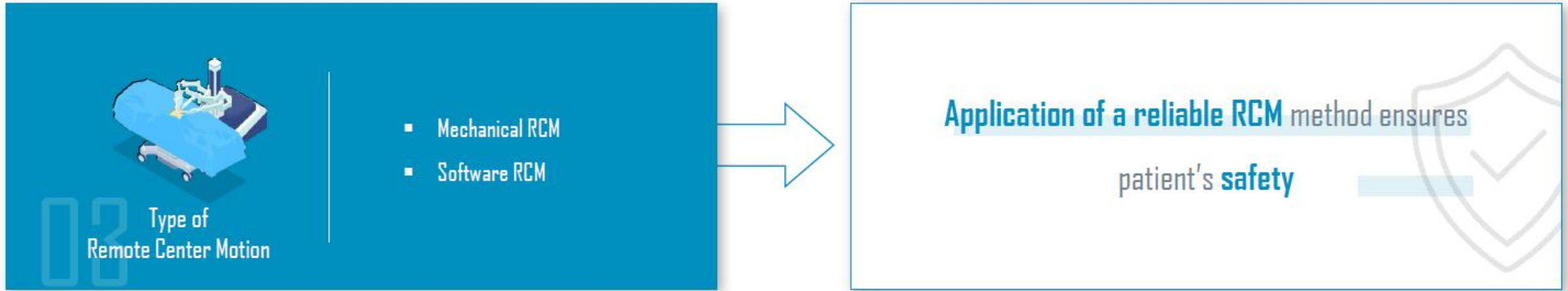


- Drape easily at designated site
- Does not occupy much space to support operation

Standalone Type

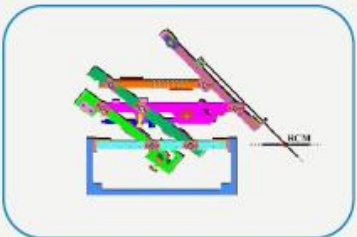


- Surgically prep and position each arm individually
- Presenting a tight workspace
- Different surgical procedures need for different arm placement guide
(Distance, Location, Line connection)



- ✓ Remote Center of Motion(RCM) :
The point at which the surgical tool is stationary provides a solid basis for movement

Mechanical RCM



✓ High-reliability applications



Software RCM



- ✓ Software algorithms to generate a virtual RCM
- ✓ It can be simplified in the robot design, but it is difficult to guarantee safety in the case of an electronic component or power malfunction
(Nisar et al J Mech Robot 2017)





Revo-i

Standalone

Mechanical RCM

Finger Gripper

Closed Console



Davinci

Standalone

Mechanical RCM

Finger Gripper

Closed Console



Versius

Modular

Software RCM

Joystick

Open Console



Hugo

Modular

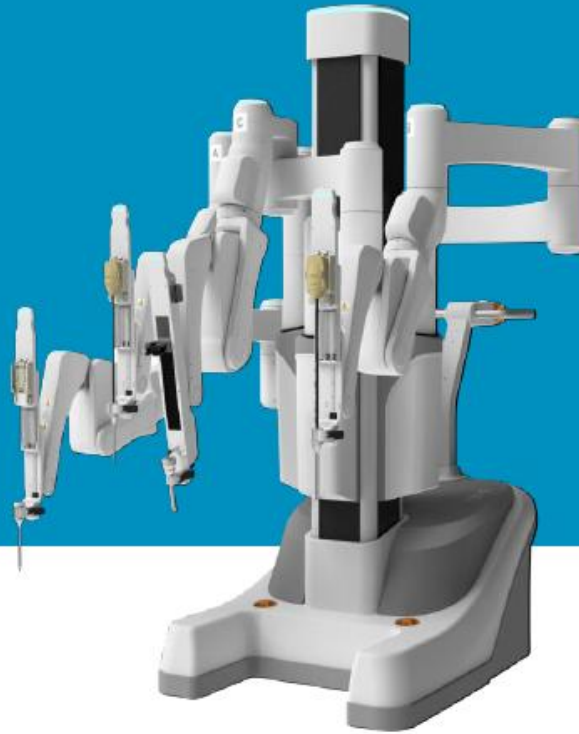
Software RCM

Pistol type

Open Console



Master Console



Operation Cart



Vision Cart

2 Revo-i MASTER CONSOLE

The Revolutionary Surgeons in Control

Ergonomically designed Revo-i™ surgical robot creates an environment where operating surgeons can truly focus on surgery

01



Feel the
3D and depth
with your own eyes.

02



Experience the natural movement
of instrument as you move the
joints
of your hand

03



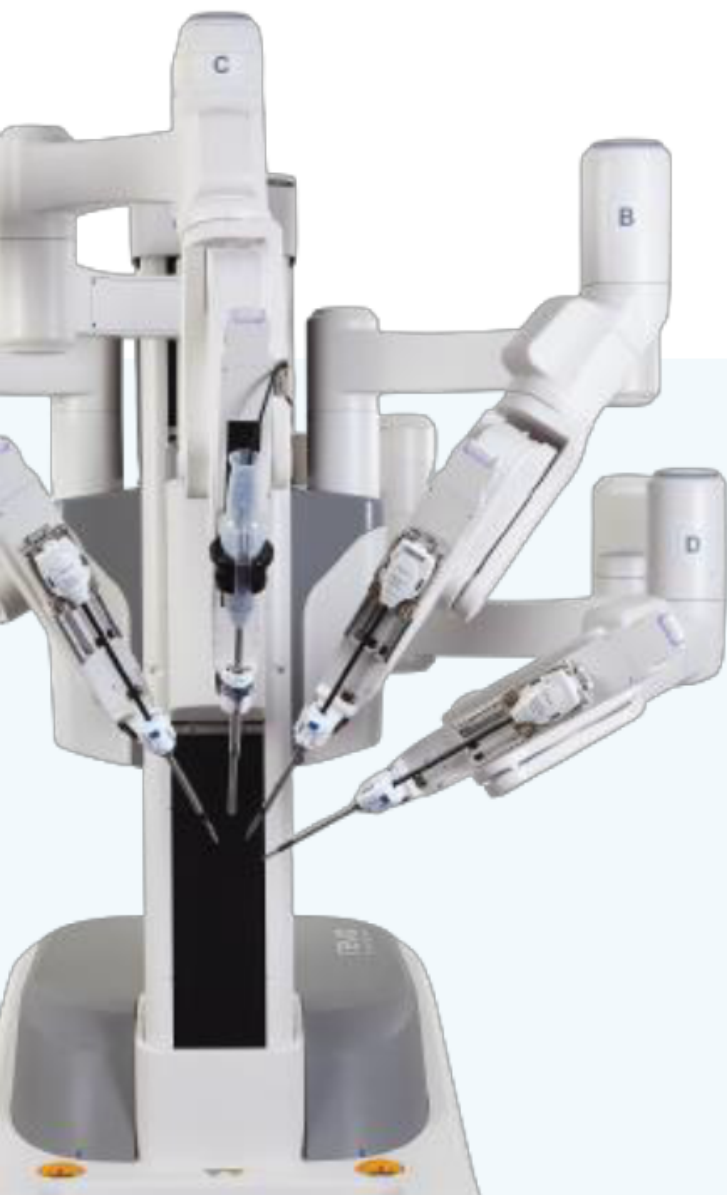
Adjustable setting for the
console head, armrest, foot
pedal

04



4 types pedal
Camera clutch pedal
Arm clutch pedal
Arm switching pedal
Energy control pedal





Seamless Motion with Articulated Revo-i™ Instruments

You can easily control the zoom-in/out function of instruments and a camera with 7 DOF(Degree of Freedom).

01



Fast recovery & less pain through RCM

RCM technology minimizes pressure on the patient's surgical site

Reducing pain and recovery time at the incision site

Instinctive movement for camera

Hold the 10.3mm endoscope firmly and stably perform surgery without tremors

02



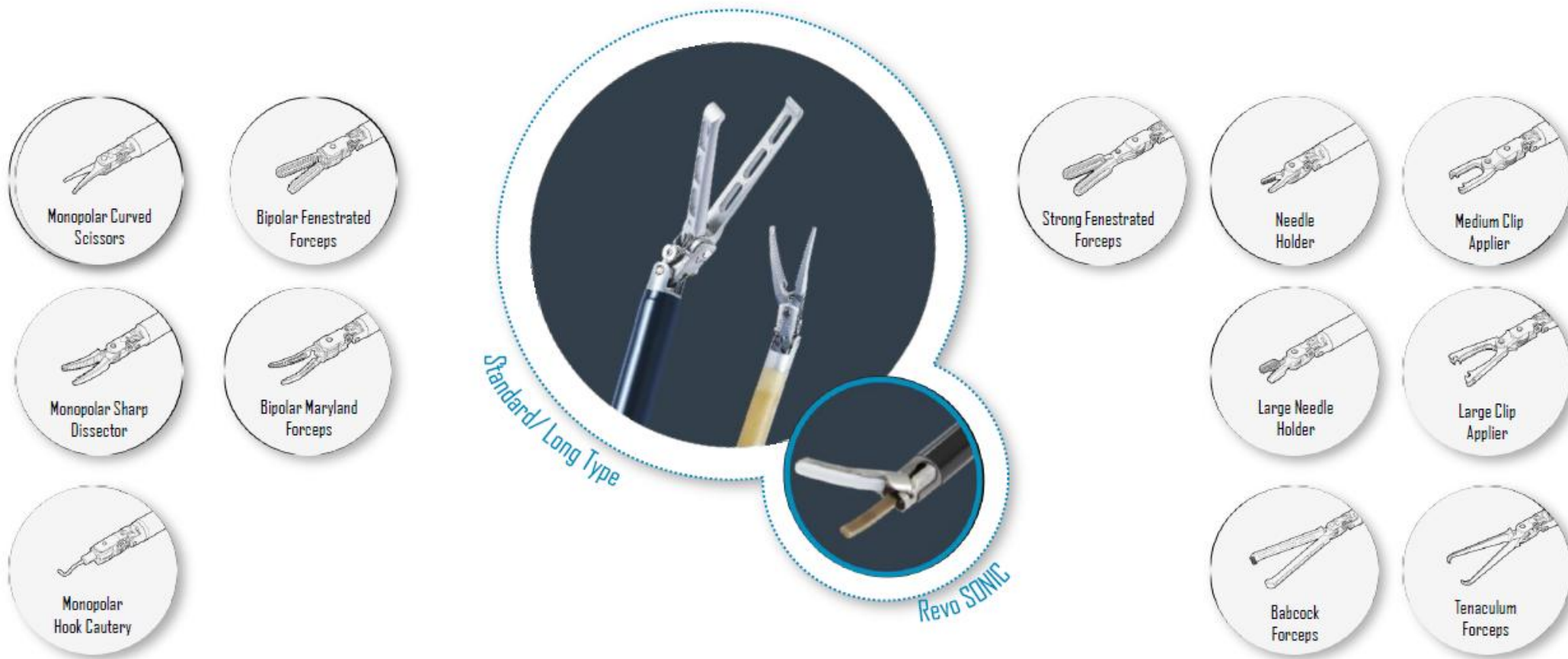
03



Reachable anatomy and precise surgery

Articulated instruments with 3 robotic arms

Seamless motion with revo-i instruments



Accessible anatomy for more precise surgery

Our multi-joint instruments with the seven degrees of freedom easily replicate your wrist movements and will allow you to operate with detailed precision during dissections, suturing, sealing, as well as retracting even in narrow surgical areas.

Vivid Visualization

You can improve the quality of minimally invasive surgery by delicately distinguishing tissues and blood vessels through high-definition magnified endoscope.



01



3D endoscope from STORZ

in Germany is linked to Revo-i system Field of view:

82 degree

- Focus-free
- Easy to attach and detach with its light weight of 420g



02



2007 - 2014

Developed 1st ~ 4th version prototype

2015 ~ 2016

Started Human Clinical Trials
Completed Pre-Clinical Trials

2017

Completed Human Clinical Trials
Acquired of manufacture approval



2018

Launched 'Revo-i'
First installation of Revo-i at Gibbeum Hospital

2020

Installed at Queens Park Women's Hospital
Installed at Severance Training Center

2021

Selected the Innovative Medical Device by MFDS
Installed at Korea Cancer Center Hospital





Revo-i case observation program at Korea Cancer Center Hospital (KCCCH)

Surgeons from abroad, including Turkey, Uzbekistan, Morocco, and Indonesia, observe Revo-i robotic surgery cases and get insights on operation of surgical robots





Ist Installation of Revo-i at Shox International Hospital in Central Asia

- ✓ Launching Ceremony for Revo-i system
- ✓ Minister of MOH announced that "The introduction of Revo-i is a cornerstone and monumental event for robotic surgery in Central Asia"
- ✓ 20+ cases has been performed by Revo-i systems within the month

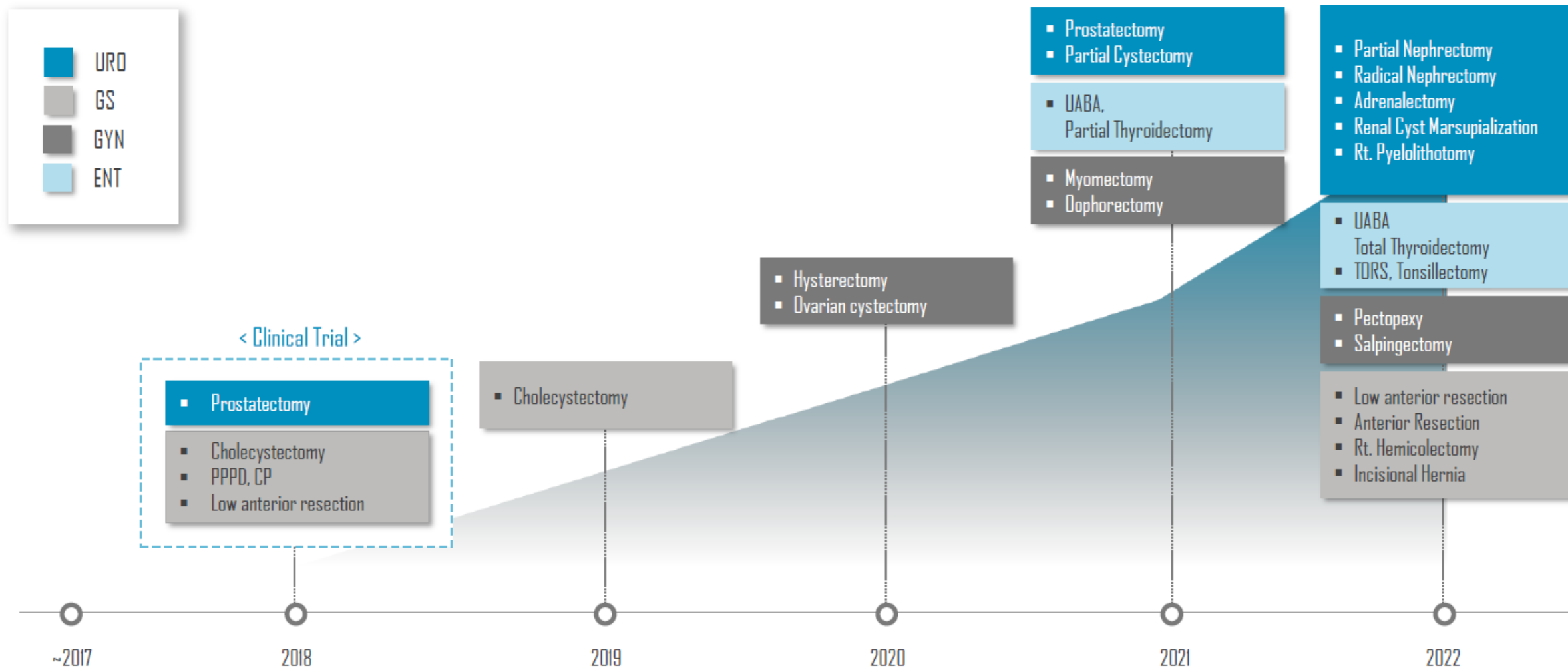


New installation at Severance hospital

- ✓ Severance Hospital signed a cooperation agreement for education and training support center, in May 2022
- ✓ Developing and carrying out wet-lab education and training programs
- ✓ We successfully installed a Revo-i at Severance Hospital in Mar 2023 for expansion of surgical application and development of K-Surgical robot

7 Revo-i EXPANSION OF REVO-I CLINICAL APPLICATION

revo-i



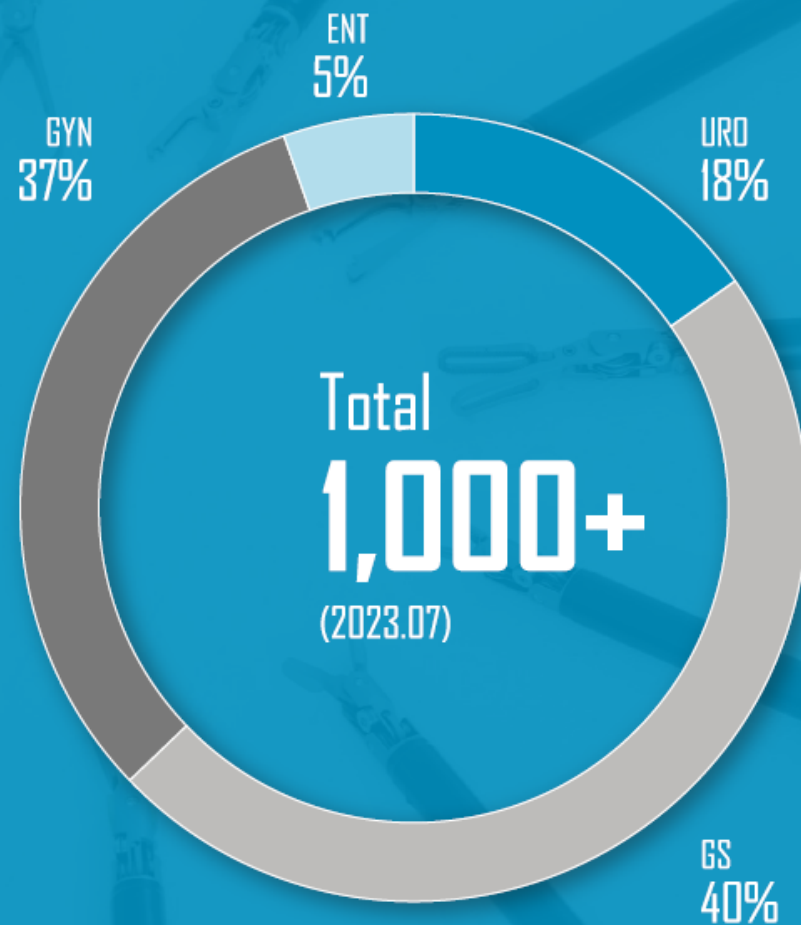


Revo-i

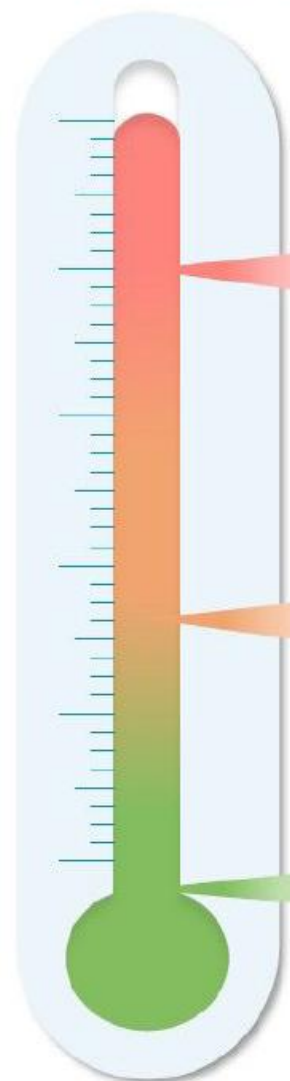
REVO-I CLINICAL CASE

Specialty	Surgery
URO	Prostatectomy
	Partial Nephrectomy
	Radical Nephrectomy
	Partial Cystectomy
	Adrenalectomy
	Renal Cyst Marsupialization
	Pyelolithotomy
GS	Low anterior resection
	Anterior Resection
	Rt. Hemicolectomy
	Cholecystectomy
	PPPD,CP
	Incisional Hernia
GYN	Myomectomy
	Hysterectomy
	Ovarian cystectomy
	Oophorectomy
	Salpingectomy
	Pectopexy
ENT	UABA, Thyroid lobectomy
	UABA, Total Thyroidectomy
	TORS, Tonsilectomy

revo-i

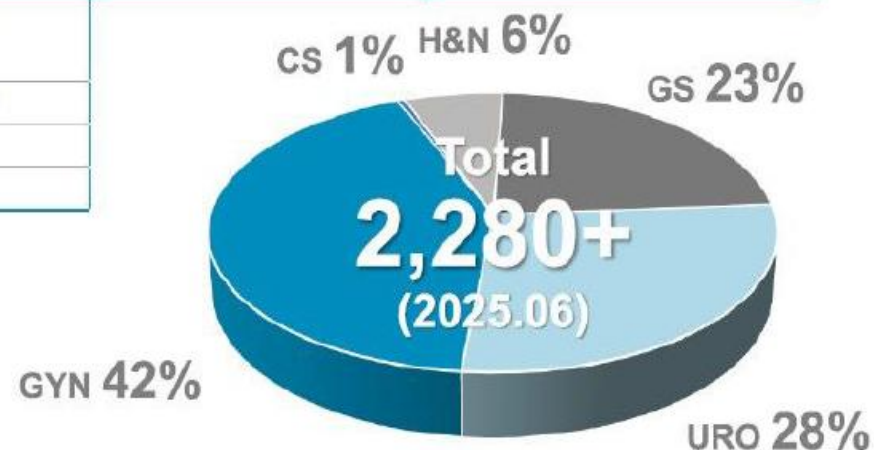


COMPLEXITY OF revo-i SURGERIES

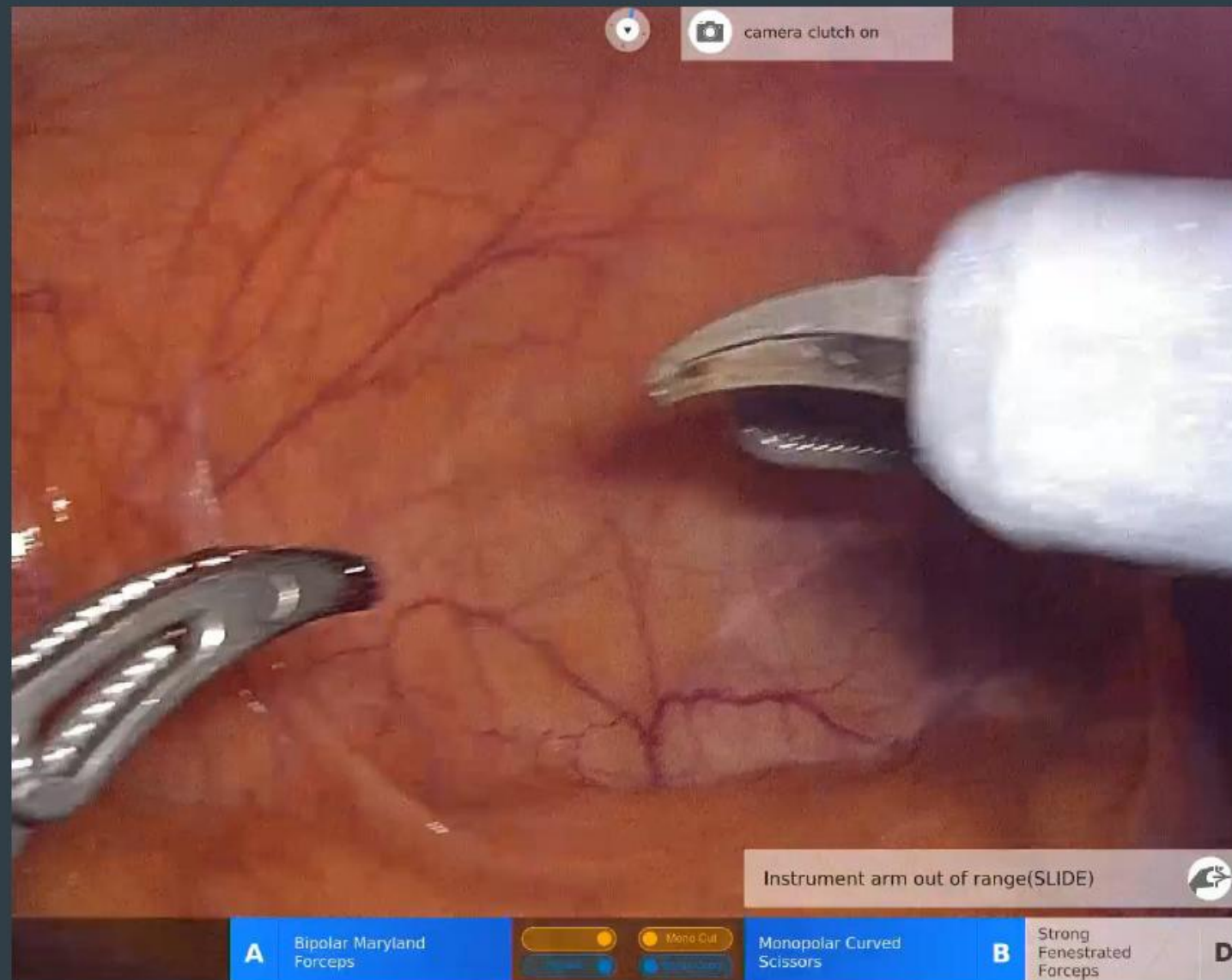


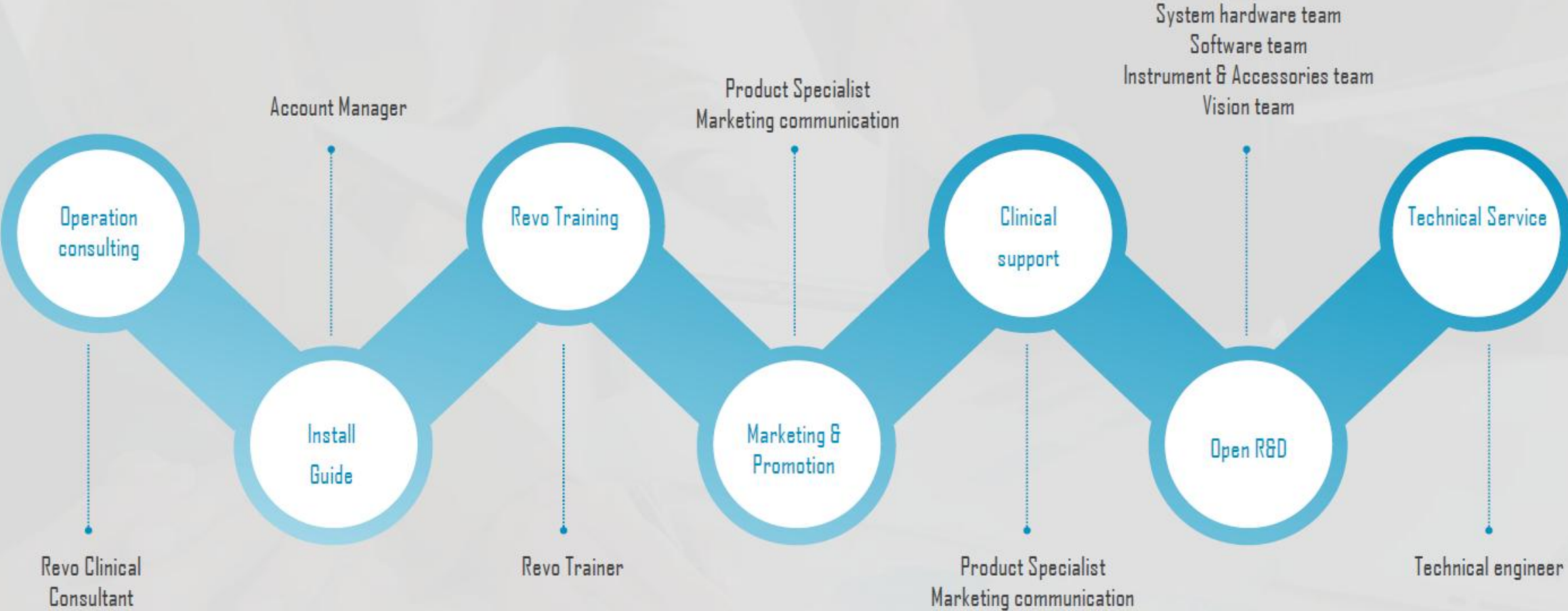
Complexity	URO	GS	Surgery GY	H&N	CS
High	Radical Cystectomy	PPPD, Whipple's op	Radical Hysterectomy	TA Thyroidectomy	Lobectomy
	Radical Prostatectomy	Total Gastrectomy	Myomectomy	BABA Thyroidectomy	
	Nephroureterectomy	Extended Cholecystectomy	Pectopexy	UABA Thyroidectomy	
	Partial Nephrectomy	Low Anterior Resection			
	Radical Nephrectomy	Abdominoperineal Resection			
Medium	Total Nephrectomy	Hepatectomy	Hysterectomy	TORS Tonsillectomy	Mediastinal Mass Removal
	Simple Prostatectomy	Hemicolectomy	Ovarian cystectomy		Hiatal Hernia Repair
	Adrenalectomy	Distal Pancreatectomy			
	Pyeloplasty	Subtotal Gastrectomy			
Low		Anterior Resection			
	Renal Cyst Marsupialization	Wedge Resection	Oophrectomy		
		Cholecystectomy	Salpingectomy		
		Hernia Repair			
		Appendectomy			

*From Simple to Complex,
Designed for Every Surgical Area*



Case: Radical Prostatectomy / M.D. KH Song. (Korea Cancer Center Hospital)

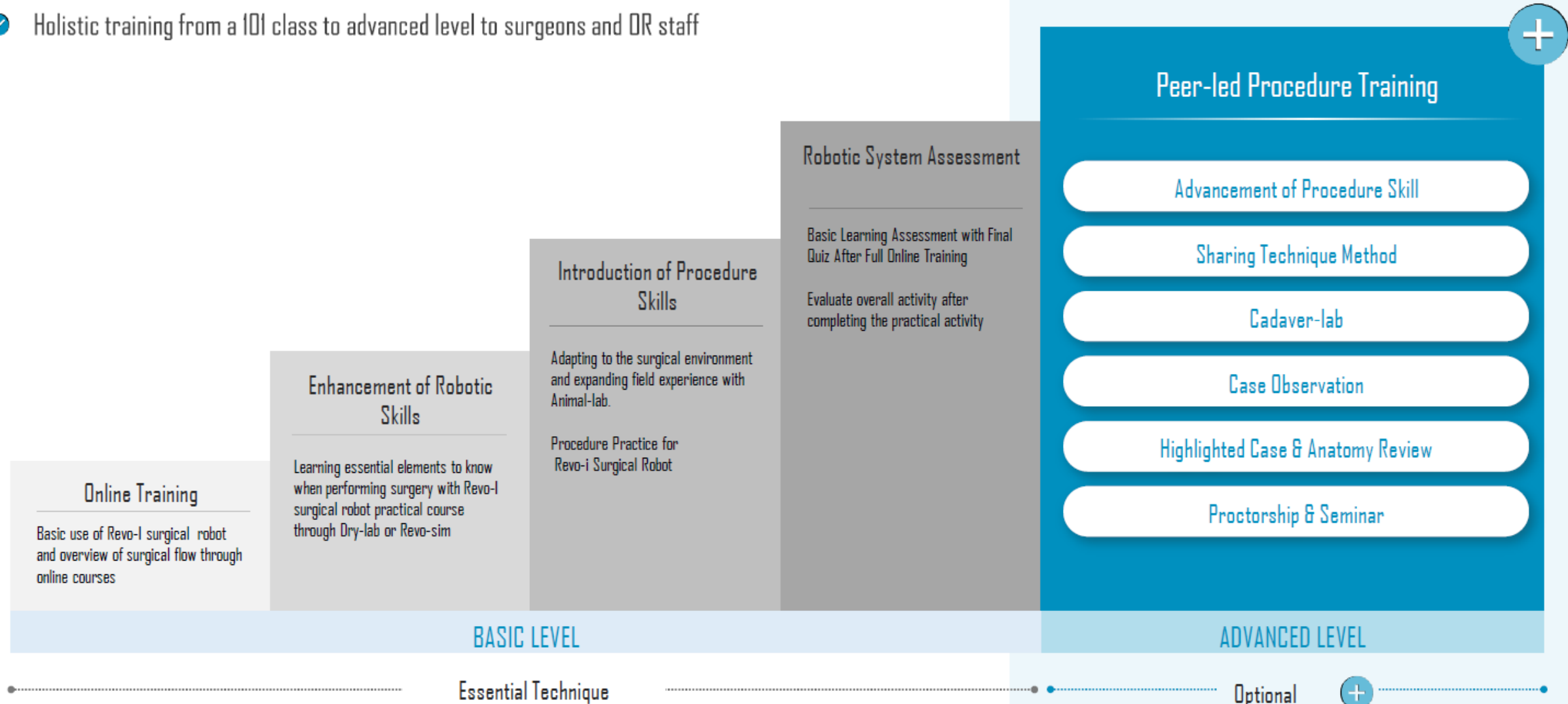




REVO TRAINING

revo-i

- ✓ Holistic training from a 101 class to advanced level to surgeons and OR staff



REVO ADVANCED TRAINING WITH SEVERANCE

revo-i

- ✓ Revo-i installed in Severance Robot and MIS Center
- ✓ meerecomapny and Severance hospital signed an agreement for tailor made Revo Training Program
- ✓ Training program includes Animal Lab, Clinical Cadaver Lab, Case Observation, Proctorship etc. based on trainee's request

SEVERANCE ROBOT AND MIS CENTER



We don't expect surgery to change to use the equipment,
we expect to **change our equipment to suit surgery**



Robotic Clinical Skill in the world



Open R&D

- System improvement
- Developing new instruments
- Procedure specific instruments



Cooperation for Robotic Surgery

- Research new applications
- Expanding the scope of Robotic Surgery